

1 9702/1/M/J/02/Q22

Which of the following correctly defines the terms *stress*, *strain* and *Young modulus*?

	stress	strain	Young modulus
A	(force) x (area)	(extension) x (original length)	(stress) / (strain)
B	(force) x (area)	(extension) / (original length)	(stress) x (strain)
C	(force) / (area)	(extension) / (original length)	(stress) / (strain)
D	(force) / (area)	(extension) x (original length)	(stress) x (strain)

2 9702/1/M/J/02/Q23

A wire is stretched by 8 mm when a load of 60 N is applied.

What will be the extension of a wire of the same material having four times the cross-sectional area and twice the original length, when the same load is applied?

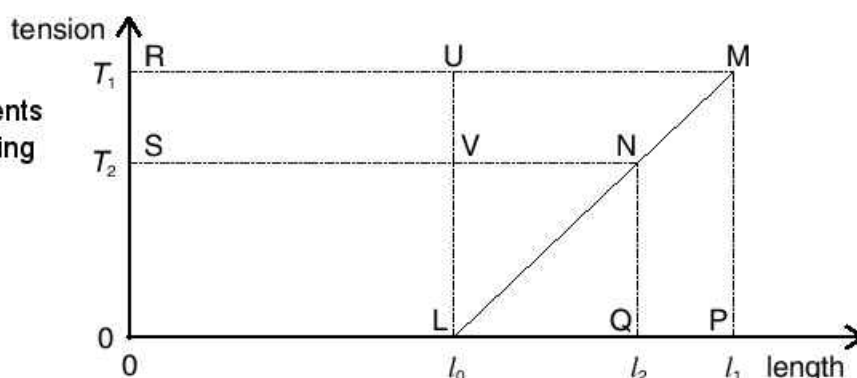
- A 2 mm B 4 mm C 8 mm D 16 mm

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The tension in a spring of natural length l_0 is first increased from zero to T_1 , causing the length to increase to l_1 . The tension is then reduced to T_2 , causing the length to decrease to l_2 (as shown).

Which area of the graph represents the work done by the spring during this reduction in length?

- A MLP B MNQP
C MNSR D MPLU



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Comparing the properties of solids, liquids and gases, which option is correct?

	property	solids	liquids	gases
A	ordering of molecules	high	not so high	random
B	spacing of molecules	close	far	far
C	translation of molecules	no	no	yes
D	vibration of molecules	no	yes	yes

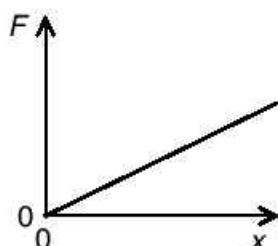
5 9702/01/O/N/03/Q21

What is the Young modulus of a metal?

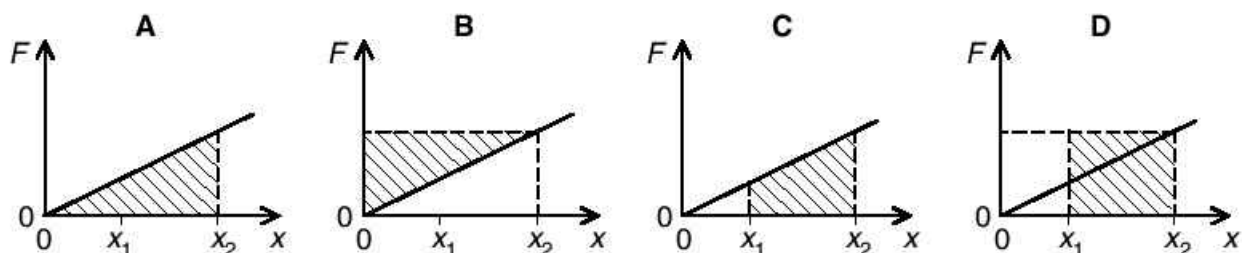
- A extension / force C strain / stress
B force / extension D stress / strain

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The variation of the extension x of a spring with applied force F is shown.



Which shaded area represents the work done when the extension is increased from x_1 to x_2 ?



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Two springs P and Q both obey Hooke's law. They have spring constants $2k$ and k respectively.

The springs are stretched, separately, by a force that is gradually increased from zero up to a certain maximum value, the same for each spring. The work done in stretching spring P is W_P , and the work done in stretching spring Q is W_Q .

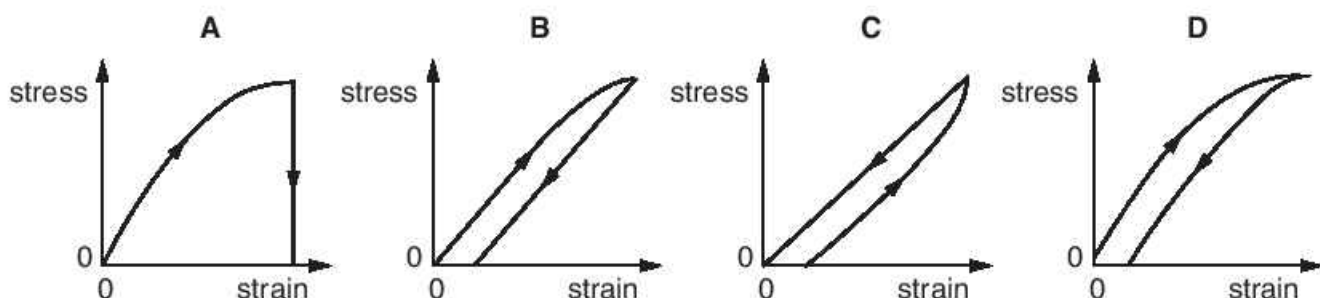
How is W_P related to W_Q ?

- A $W_P = \frac{1}{4}W_Q$ B $W_P = \frac{1}{2}W_Q$ C $W_P = 2W_Q$ D $W_P = 4W_Q$

8 9702/01/M/J/03/Q19

A suspended copper wire is gradually loaded until it is stretched just beyond the elastic limit, and it is then gradually unloaded.

Which graph (with arrows indicating the sequence) best illustrates the variation of the tensile stress with longitudinal strain?



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What is the ultimate tensile stress of a material?

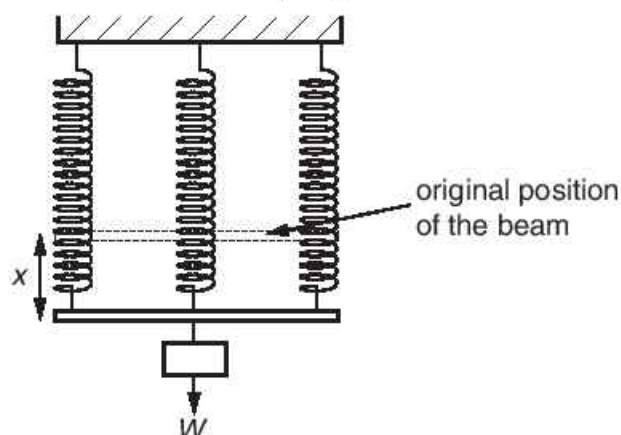
- A the stress at which the material becomes ductile
B the stress at which the material breaks
C the stress at which the material deforms plastically
D the stress at which the material reaches its elastic limit

10 9702/01/M/J/03/Q22

A beam, the weight of which may be neglected, is supported by three identical springs. When a weight W is hung from the middle of the beam, the extension of each spring is x .

The middle spring and the weight are removed.
What is the extension when a weight of $2W$ is hung from the middle of the beam?

- A $\frac{3x}{2}$ B $\frac{4x}{3}$
C $2x$ D $3x$

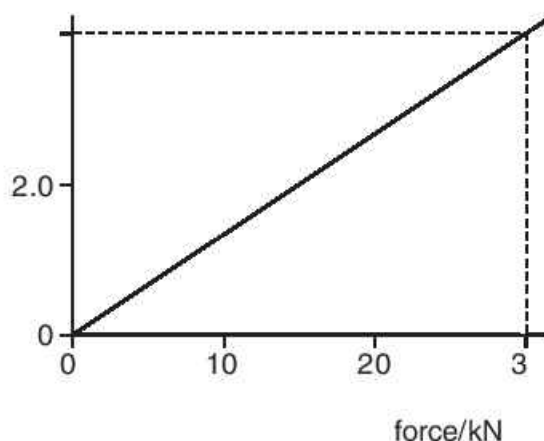


11 9702/01/O/N/03/Q22

The graph shows how the extension of a spring varies with the force used to stretch it.

What is the strain energy stored in the spring when the extension is 4.0 cm?

- A 60 J B 120 J
C 600 J D 1200 J



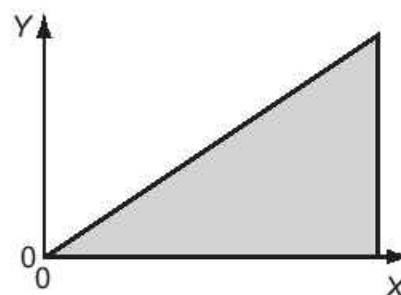
12 9702/01/M/J/04/Q22

The graph shown was plotted in an experiment on a metal wire.

The shaded area represents the total strain energy stored in stretching the wire.

How should the axes be labelled?

	Y	X
A	force	extension
B	mass	extension
C	strain	energy
D	stress	strain



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Nylon breaks when the stress within it reaches 1×10^9 Pa.

Which range includes the heaviest load that could be lifted by a nylon thread of diameter 1 mm?

- A 2 N to 20 N C 200 N to 2000 N
B 20 N to 200 N D 2000 N to 20 000 N

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The table shows a load applied to four wires and the cross-sectional area of each. Which of the wires is subjected to the greatest stress?

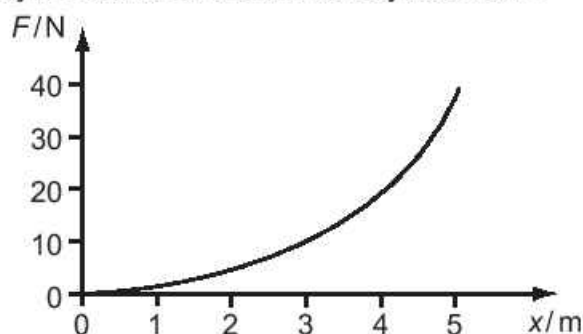
	load / N	cross-sectional area / mm ²
A	1500	0.25
B	2000	1.0
C	3000	0.56
D	5000	2.3

15 9702/01/O/N/04/Q23

The force F required to extend a sample of rubber by a distance x is found to vary as shown.

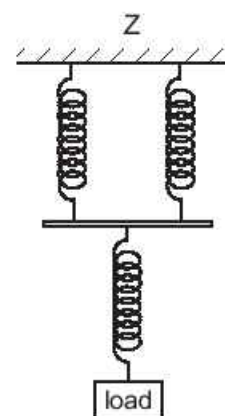
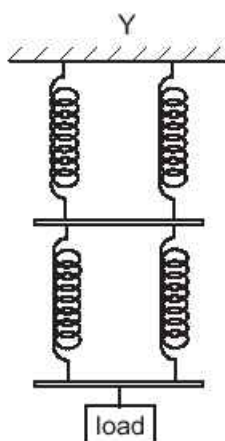
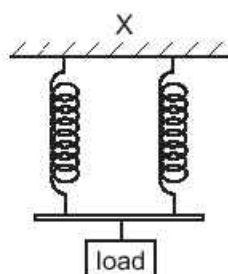
The energy stored in the rubber for an extension of 5 m is

- A** less than 100 J.
- B** 100 J.
- C** between 100 J and 200 J.
- D** more than 200 J.



16 9702/01/M/J/05/Q20

A number of similar springs, each having the same spring constant, are joined in three arrangements X, Y and Z. The same load is applied to each.



What is the order of increasing extension for these arrangements?

	smallest	→	largest
A	X	Y	Z
B	Z	X	Y
C	Z	Y	X
D	Y	X	Z

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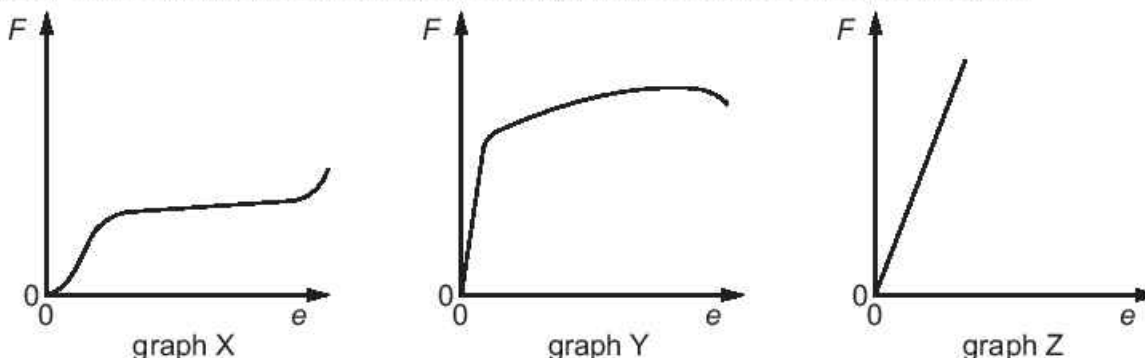
What is the unit of the Young modulus?

- A** Nm⁻¹
- B** Nm
- C** Nm⁻²
- D** Nm²

- 18 **9702/12/O/N/13/Q21***
Which two substances are normally both crystalline?

A copper and diamond C diamond and glass
B copper and glass D diamond and rubber

- 19 **9702/13/M/J/14/Q24**
Cylindrical samples of steel, glass and rubber are each subjected to a gradually increasing tensile force F . The extensions e are measured and graphs are plotted as shown below.



Which row correctly relates the graphs to the materials?

	steel	glass	rubber
A	X	Y	Z
B	X	Z	Y
C	Y	X	Z
D	Y	Z	X

- 20 **9702/01/M/J/05/Q22***
Two steel wires P and Q have lengths l and $2l$ respectively, and cross-sectional areas A and $\frac{A}{2}$ respectively. Both wires obey Hooke's law.

What is the ratio $\frac{\text{tension in P}}{\text{tension in Q}}$ when both wires are stretched to the same extension?

A $\frac{1}{4}$ B $\frac{1}{2}$ C $\frac{2}{1}$ D $\frac{4}{1}$

- 21 **9702/01/O/N/05/Q19***
When white sugar granules are heated, they melt. When the melt is cooled quickly, a brittle solid form of toffee is produced.

How does the structure of the sugar change?

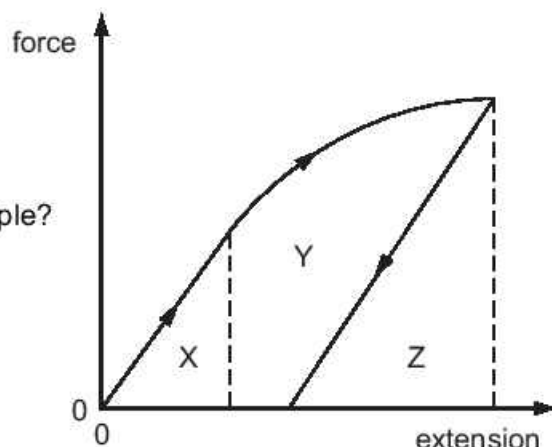
A amorphous to polymeric C crystalline to polymeric
B crystalline to amorphous D polymeric to amorphous

- 22 **9702/01/M/J/06/Q21**
In describing the behaviour of a spring, the spring constant is used. Different loads are used to extend the spring by different amounts. To find the spring constant, which quantities are required?

A the elastic limit and the loads
B the elastic limit, extensions and the length of the spring
C the loads and the extensions of the spring
D the loads and the length of the spring

23 9702/01/O/N/05/Q20

A ductile material is stretched by a tensile force to a point beyond its elastic limit. The tensile force is then reduced to zero. The graph of force against extension is shown below.



Which area represents the net work done on the sample?

- A X B X + Y
C Y + Z D Z

24 9702/01/O/N/05/Q21

A wire stretches 8 mm under a load of 60 N.

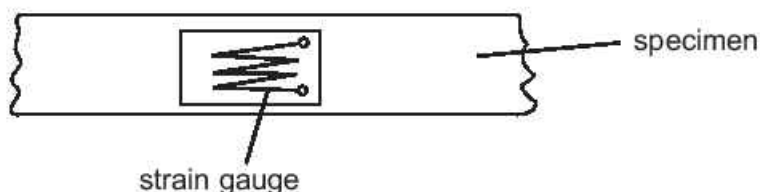
A second wire of the same material, with half the diameter and a quarter of the original length of the first wire, is stretched by the same load.

Assuming that Hooke's law is obeyed, what is the extension of this wire?

- A 1 mm B 4 mm C 8 mm D 16 mm

25 9702/01/O/N/05/Q33

Tensile strain may be measured by the change in electrical resistance of a strain gauge. A strain gauge consists of folded fine metal wire mounted on a flexible insulating backing sheet. The strain gauge is firmly attached to the specimen, so that the strain in the metal wire is always identical to that in the specimen.



When the strain in the specimen is increased, what happens to the resistance of the wire?

- A It decreases, because the length decreases and the cross-sectional area increases.
B It decreases, because the length increases and the cross-sectional area decreases.
C It increases, because the length decreases and the cross-sectional area increases.
D It increases, because the length increases and the cross-sectional area decreases.

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In an experiment to demonstrate Brownian motion, smoke particles in a container are illuminated by a strong light source and observed through a microscope.

The particles are seen as small specks of light that are in motion.

What causes the Brownian motion?

- A collisions between the smoke particles and air molecules
B collisions between the smoke particles and the walls of the container
C convection currents within the air as it is warmed by the light source
D kinetic energy gained by the smoke particles on absorption of light

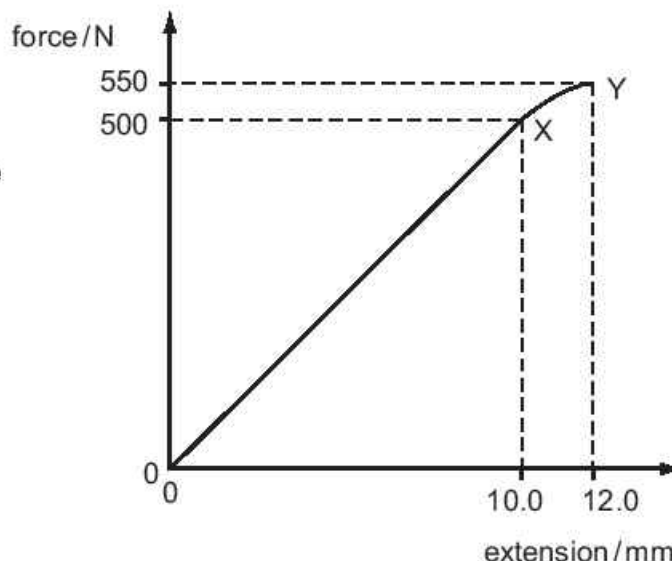
27 9702/01/M/J/06/Q22

The graph shows the behaviour of a sample of a metal when it is stretched until it starts to undergo plastic deformation.

What is the total work done in stretching the sample from zero extension to 12.0 mm?

Simplify the calculation by treating the region XY as a straight line.

- A 3.30 J B 3.55 J
C 3.60 J D 6.60 J



28 9702/01/O/N/06/Q22

What is represented by the gradient of a graph of force (vertical axis) against extension (horizontal axis)?

- A elastic limit B spring constant C stress D the Young modulus

29 9702/01/M/J/07/Q17

A piece of copper is drawn into a continuous wire.

What behaviour is the copper exhibiting?

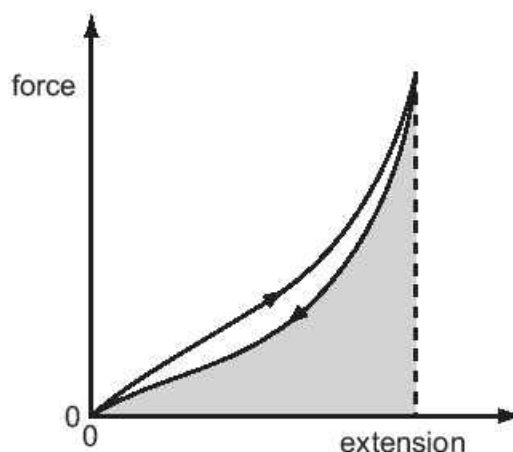
- A brittle only B elastic only C plastic only D both brittle and elastic

30 9702/01/M/J/07/Q18

The force-extension graph of a particular sample of rubber as a load is applied and then removed is shown.

What does the shaded area represent?

- A the energy transformed into heat during the complete cycle
B the recoverable elastic potential energy stored at maximum extension
C the work done on the sample while loading
D the work done on the sample while unloading



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Which properties best describe modelling clay?

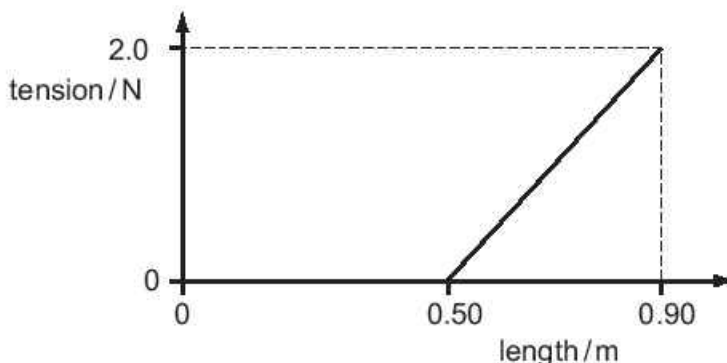
- A brittle and ductile C elastic and plastic
B ductile and elastic D plastic and ductile

32 9702/01/M/J/07/Q19

A spring of unextended length 0.50 m is stretched by a force of 2.0 N to a new length of 0.90 m. The variation of its length with tension is as shown.

How much strain energy is stored in the spring?

- A 0.40 J B 0.80 J
C 0.90 J D 1.8 J



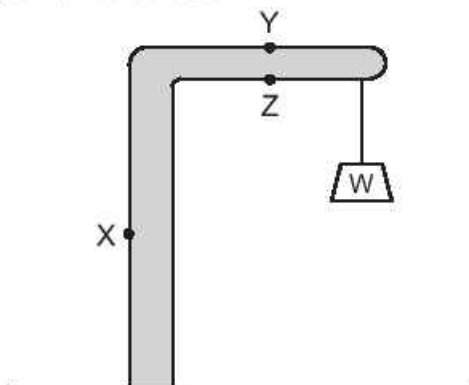
33 9702/01/M/J/07/Q20

A simple crane consists of a rigid vertical pillar supporting a horizontal beam.

A weight W is lifted by a rope at the end of the beam.

What are the forces at points X, Y and Z due to the weight W ?

	force at X	force at Y	force at Z
A	tension	compression	tension
B	tension	tension	compression
C	compression	tension	compression
D	compression	compression	compression



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What is plastic deformation?

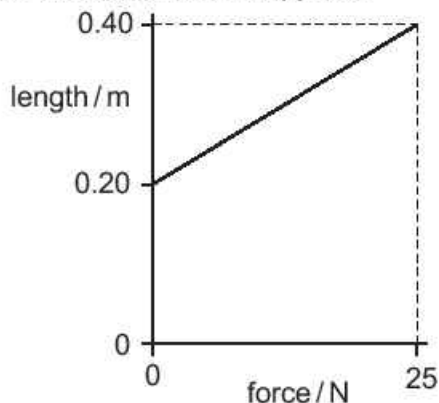
- A Plastic deformation occurs when strain is not proportional to stress but when the load is removed the material returns to its original length.
B Plastic deformation occurs if, when the load is removed, the material contracts but a permanent stretching has occurred.
C Plastic deformation occurs until the extension is no longer proportional to the load.
D Plastic deformation occurs when the material extends so that strain is directly proportional to stress.

35 9702/01/O/N/07/Q20

The graph shows how the length of a particular rubber cord varies as force is applied.

What is the maximum strain energy in this deformed rubber cord?

- A 2.5 J
B 5.0 J
C 7.5 J
D 10 J



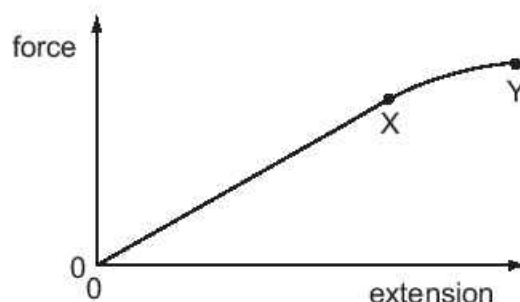
36 9702/01/M/J/08/Q22

A sample of metal is subjected to a force which increases to a maximum value and then decreases back to zero. A force-extension graph for the sample is shown.

When the sample contracts it follows the same force-extension curve as when it was being stretched.

What is the behaviour of the metal between X and Y?

- A both elastic and plastic
- B elastic but not plastic
- C plastic but not elastic
- D not elastic and not plastic

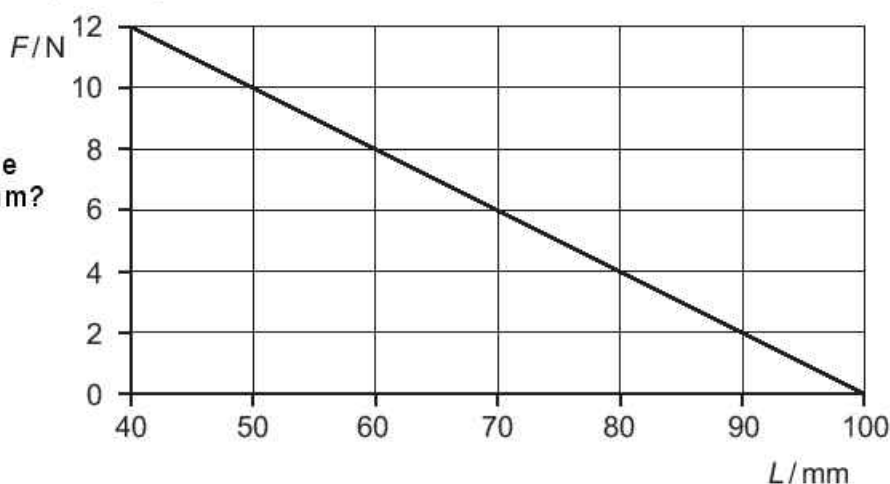


37 9702/01/M/J/08/Q23

A spring of original length 100 mm is compressed by a force. The graph shows the variation of the length L of the spring with the compressing force F .

What is the energy stored in the spring when the length is 70 mm?

- A 0.090 J B 0.21 J
- C 0.27 J D 0.63 J



38 9702/01/M/J/08/Q24

The Young modulus of steel is determined using a length of steel wire and is found to have the value E .

Another experiment is carried out using a wire of the same steel, but of twice the length and half the diameter.

What value is obtained for the Young modulus in the second experiment?

- A $\frac{1}{4}E$ B $\frac{1}{2}E$ C E D $2E$

39 9702/01/O/N/08/Q23

For a wire, Hooke's law is obeyed for a tension F and extension x . The Young modulus for the material of the wire is E .

Which expression represents the elastic strain energy stored in the wire?

- A $\frac{1}{2}Ex$ B Ex C $\frac{1}{2}Fx$ D Fx

40 9702/01/M/J/09/Q19

Four materials are formed into rods of the same dimensions.

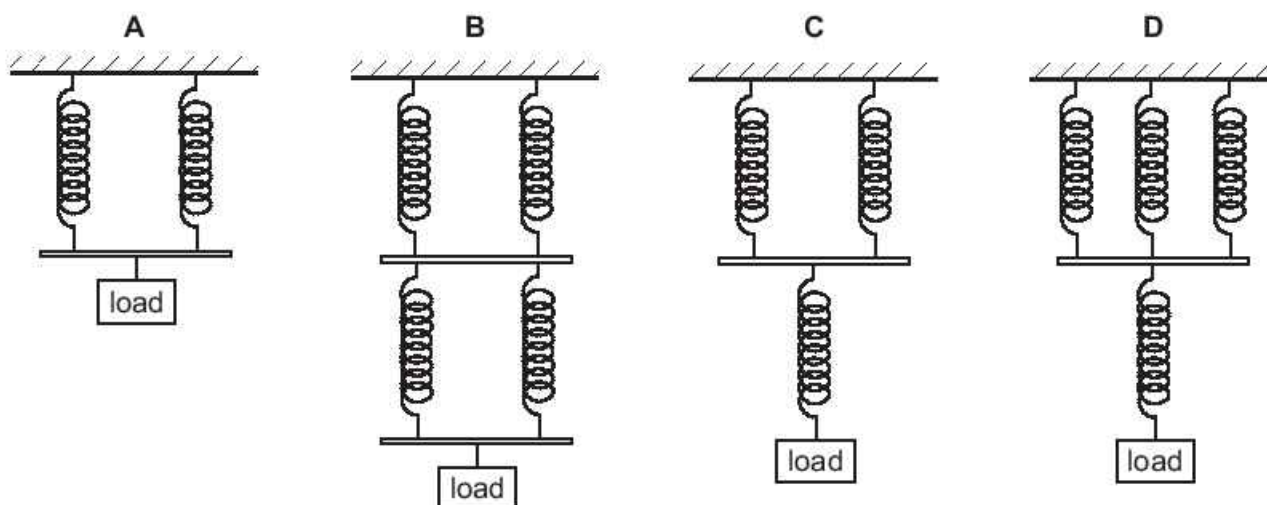
At room temperature, which can sustain the largest plastic deformation?

- A the ductile material aluminium C the brittle material glass
- B the brittle material carbon D the ductile material steel

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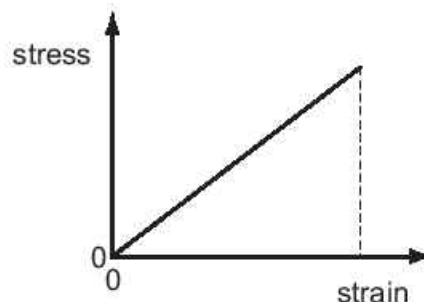
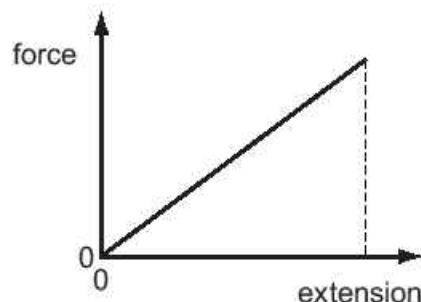
A number of similar springs, each having the same spring constant, are joined in four arrangements. The same load is applied to each.

Which arrangement gives the greatest extension?



42 9702/01/O/N/08/Q22

The graphs show how force varies with extension and stress varies with strain for the loading of a metal wire.



The Young modulus for this wire is equal to

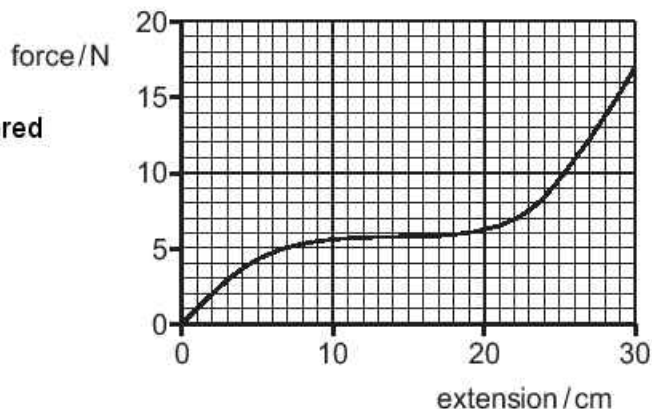
- A the gradient of the force-extension graph.
- B the area between the force-extension graph and the extension axis.
- C the gradient of the stress-strain graph.
- D the area between the stress-strain graph and the strain axis.

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A rubber band is stretched by hanging weights on it and the force-extension graph is plotted from the results.

What is the best estimate of the strain energy stored in the rubber band when it is extended 30 cm?

- A 2.0 J B 2.6 J
- C 5.1 J D 200 J



44 9702/01/M/J/09/Q20

Two steel wires P and Q have lengths l and $2l$ respectively, and cross-sectional areas A and $\frac{A}{2}$ respectively. Both wires obey Hooke's law.

What is the ratio $\frac{\text{tension in P}}{\text{tension in Q}}$ when both wires are stretched to the same extension?

- A $\frac{1}{4}$ B $\frac{1}{2}$ C $\frac{2}{1}$ D $\frac{4}{1}$

45 9702/12/O/N/09/Q19

Which row best defines elastic and plastic behaviour of a material?

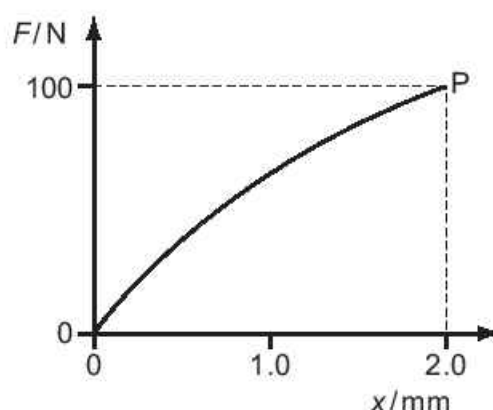
	elastic behaviour of a material	plastic behaviour of a material
A	extends only within the limit of proportionality	extends beyond the limit of proportionality
B	has a linear force-extension curve	has a horizontal force-extension curve
C	obeys Hooke's Law	extends continuously under a steady load
D	returns to its original shape and size	suffers permanent deformation

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The graph shows the non-linear force-extension curve for a wire made from a new composite material.

What could be the value of the strain energy stored in the wire when it is stretched to point P?

- A 0.09 J B 0.10 J
C 0.11 J D 0.20 J



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A steel string on an electric guitar has the following properties.

diameter = 5.0×10^{-4} m

Young modulus = 2.0×10^{11} Pa

tension = 20 N

The string snaps, and contracts elastically.

By what percentage does a length l of a piece of the string contract?

- A $5.1 \times 10^{-4} \%$ B $5.1 \times 10^{-2} \%$ C $1.3 \times 10^{-4} \%$ D $1.3 \times 10^{-2} \%$

48 9702/11/O/N/10/Q20

A student writes some statements about solids, liquids and gases.

- 1 Solids are rigid because the molecules in a solid vibrate.
- 2 Liquids flow because the molecules in a liquid are closer than in a gas.
- 3 Gases are less dense than liquids because the molecules in a gas move randomly.

Which statements are correct?

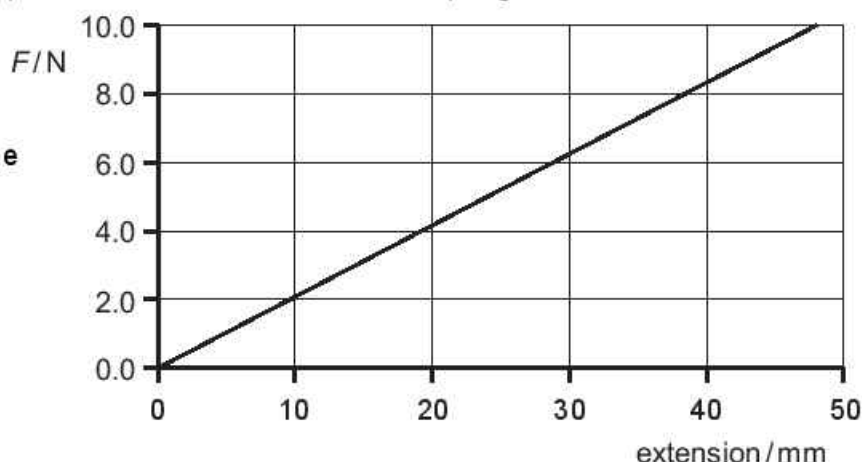
- A 1 only B 1 and 3 only C 2 and 3 only D none of the above

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The graph shows how force depends on extension for a certain spring.

What is the energy stored in the spring when the extension is 30 mm?

- A 0.095 J B 0.19 J
C 0.25 J D 0.95 J



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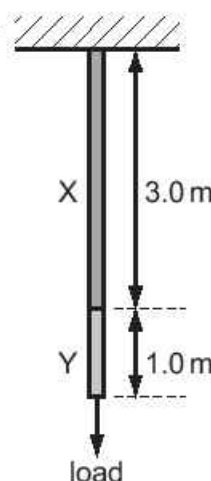
A wire consists of a 3.0 m length of metal X joined to a 1.0 m length of metal Y. The cross-sectional area of the wire is uniform.

A load hung from the wire causes metal X to stretch by 1.5 mm and metal Y to stretch by 1.0 mm.

The same load is then hung from a second wire of the same cross-section, consisting of 1.0 m of metal X and 3.0 m of metal Y.

What is the total extension of this second wire?

- A 2.5 mm B 3.5 mm C 4.8 mm D 5.0 mm



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Which group of materials contains two polymers?

- | | | | | | | | |
|---|--------|------|-------------|---|-------|---------|--------|
| A | copper | sand | polystyrene | C | nylon | sugar | rubber |
| B | glass | wood | aluminium | D | stone | diamond | steel |

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Two wires P and Q are made from the same material.

Wire P is initially twice the diameter and twice the length of wire Q. The same force, applied to each wire, causes the wires to extend elastically.

What is the ratio of the extension in P to that in Q?

- A $\frac{1}{2}$ B 1 C 2 D 4

53 9702/11/O/N/11/Q23

The Young modulus of steel is determined using a length of steel wire and is found to have the value E .

Another experiment is carried out using a wire of the same steel, but of half the length and half the diameter.

What value is obtained for the Young modulus in the second experiment?

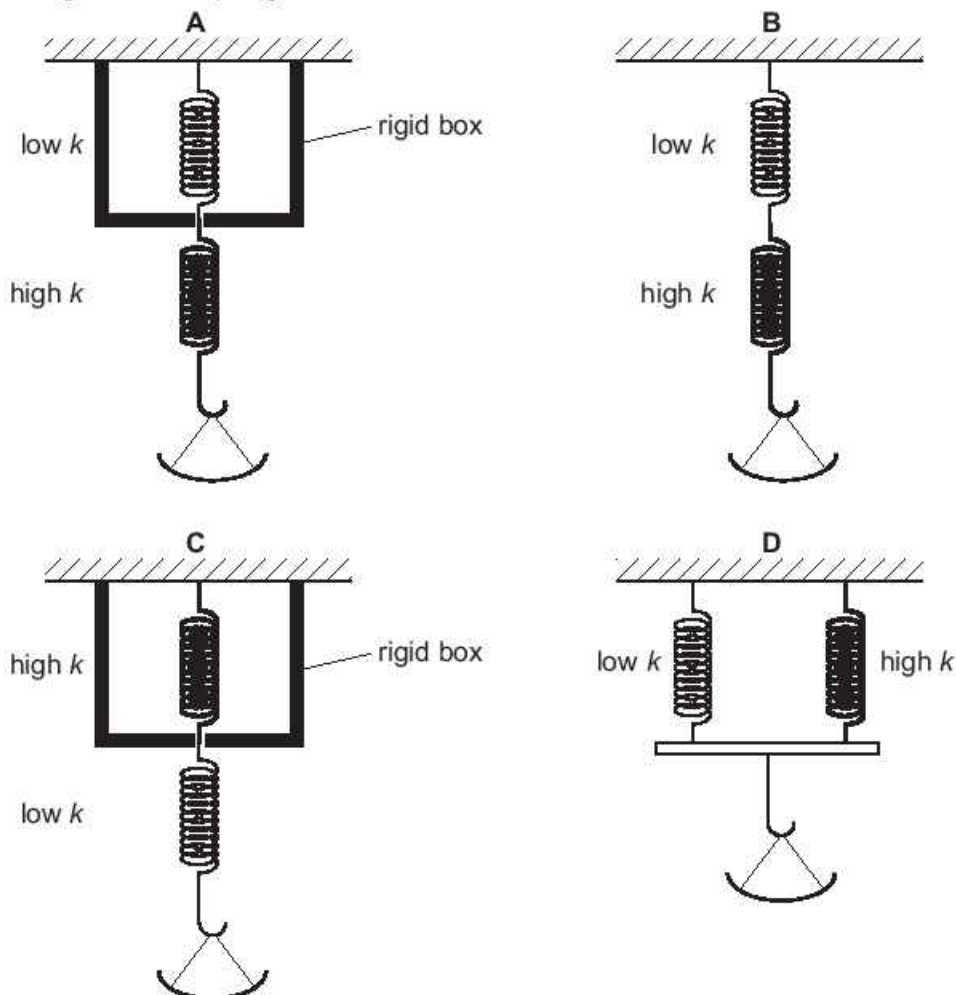
- A $\frac{1}{2}E$ B E C $2E$ D $4E$

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To determine the mass of food in a pan, a scale is used that has high sensitivity for small masses but low sensitivity for large masses.

To do this, two springs are used, each with a different spring constant k . One of the springs has a low spring constant and the other has a high spring constant.

Which arrangement of springs would be suitable?



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The Young modulus E can be determined from measurements made when a wire is stretched.

Which quantities would be measured in order to determine E ?

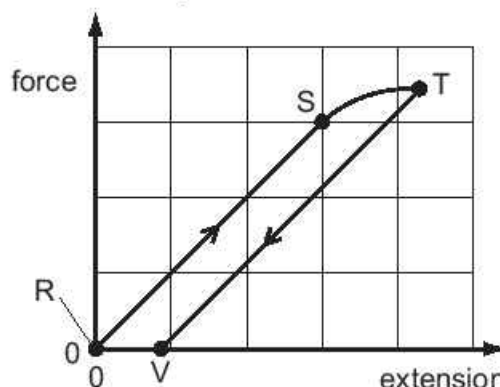
A	mass of stretching load	original length of wire	diameter of wire	extension of wire
B	mass of stretching load	new length of wire	cross-sectional area of wire	diameter of wire
C	mass of wire	original length of wire	cross-sectional area of wire	new length of wire
D	mass of wire	new length of wire	diameter of wire	extension of wire

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A long, thin metal wire is suspended from a fixed support and hangs vertically. Masses are suspended from its lower end.

The load on the lower end is increased from zero and then decreased again back to zero.

The diagram shows the force-extension graph produced.



Where on the graph would the elastic limit be found?

- A anywhere between point R and point S
- B beyond point S but before point T
- C exactly at point S
- D exactly at point T

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The behaviour of a wire under tensile stress may be described in terms of the Young modulus E of the material of the wire and of the force per unit extension k of the wire.

For a wire of length L and cross-sectional area A , what is the relation between E and k ?

- A $E = \frac{A}{kL}$
- B $E = \frac{kA}{L}$
- C $E = \frac{kL}{A}$
- D $E = \frac{L}{kA}$

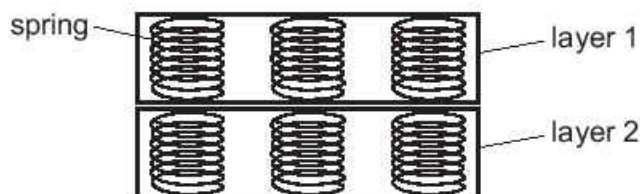
58 9702/12/M/J/11/Q24

The diagram shows the structure of part of a mattress.

The manufacturer wants to design a softer mattress (one which will compress more for the same load).

Which change will **not** have the desired effect?

- A using more layers of springs
- B using more springs per unit area
- C using springs with a smaller spring constant
- D using springs made from wire with a smaller Young modulus



59 9702/11/O/N/11/Q5

The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

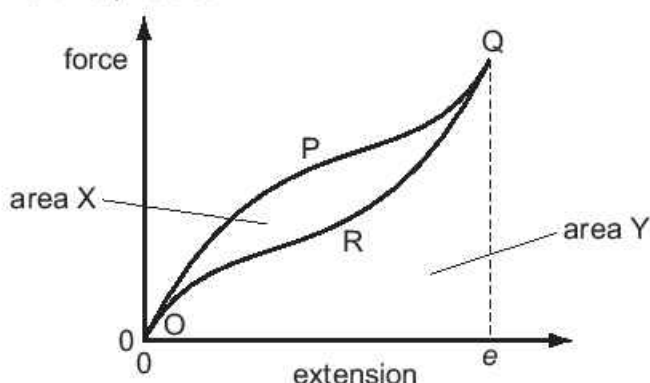
The wire is extended by a known force and the following measurements are made.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	measurement	symbol	value
A	length of wire before force applied	l	$2.043 \pm 0.002 \text{ m}$
B	diameter of wire	d	$0.54 \pm 0.02 \text{ mm}$
C	force applied	F	$19.62 \pm 0.01 \text{ N}$
D	extension of wire with force applied	x	$5.2 \pm 0.2 \text{ mm}$

60 9702/11/O/N/11/Q24

A rubber band is stretched and then relaxed to its original length. The diagram shows the force-extension graph for this process.



As the force is increased, the curve follows the path OPQ to extension e . As the force is reduced, the curve follows the path QRO to return to zero extension.

The area labelled X is between the curves OPQ and QRO. The area labelled Y is bounded by the curve QRO and the horizontal axis.

Which statement about the process is correct?

- A Area X is the energy which heats the band as it is stretched to e .
- B (Area X + area Y) is the minimum energy required to stretch the band to e .
- C Area X is the elastic potential energy stored in the band when it is stretched to e .
- D (Area Y – area X) is the net work done on the band during the process.

61 9702/11/O/N/11/Q25

When describing the behaviour of a spring, the spring constant is used.

Different loads are used to extend the spring by different amounts.

To find the spring constant, which quantities are required?

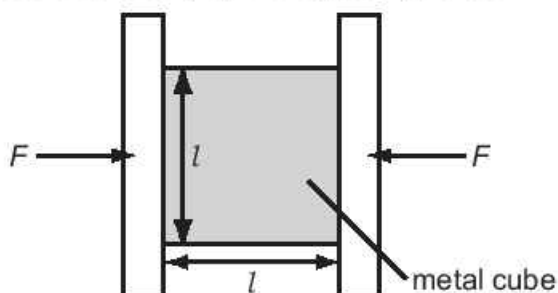
- A the elastic limit and the loads
- B the elastic limit, extensions and the length of the spring
- C the loads and the extensions of the spring
- D the loads and the length of the spring

62 9702/11/O/N/11/Q26

A metal cube of side l is placed in a vice and compressed elastically by two opposing forces F .

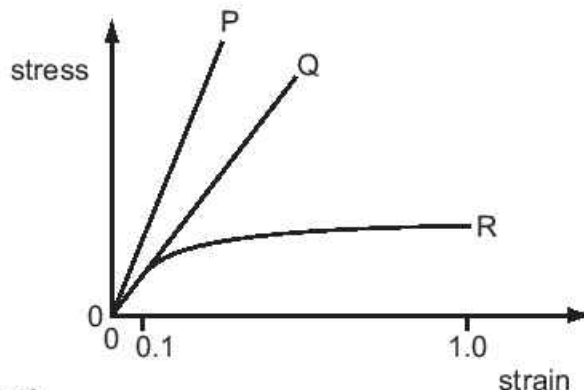
How will Δl , the amount of compression, relate to l ?

- A $\Delta l \propto \frac{1}{l^2}$
- B $\Delta l \propto \frac{1}{l}$
- C $\Delta l \propto l$
- D $\Delta l \propto l^2$



63 9702/12/O/N/11/Q21

The graph shows the relationship between stress and strain for three wires of the same linear dimensions but made from different materials.



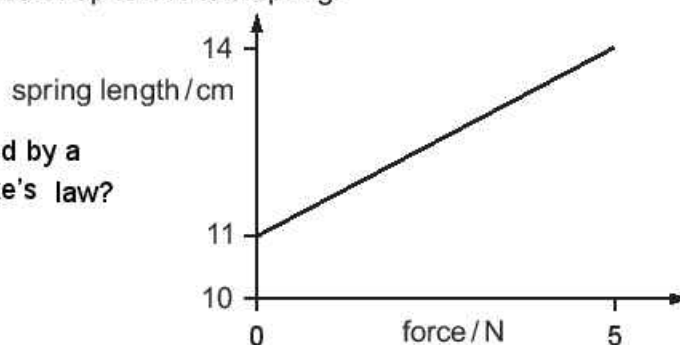
Which statements are correct?

- 1 The extension of P is approximately twice that of Q for the same stress.
- 2 The ratio of the Young modulus for P to that of Q is approximately two.
- 3 For strain less than 0.1, R obeys Hooke's law.

A 1, 2 and 3 **B** 1 and 3 only **C** 2 and 3 only **D** 2 only

64 9702/12/O/N/11/Q22

The graph shows the effect of applying a force of up to 5 N to a spring.

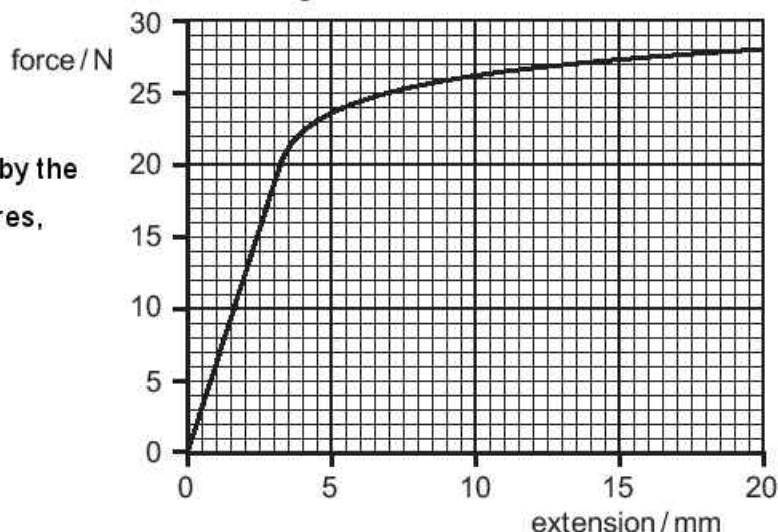


What is the total increase in length produced by a 7 N force, assuming the spring obeys Hooke's law?

- A** 4.2 cm **B** 5.6 cm
C 15.2 cm **D** 19.6 cm

65 9702/12/M/J/12/Q24

The graph is a force-extension graph for a wire that is being stretched.



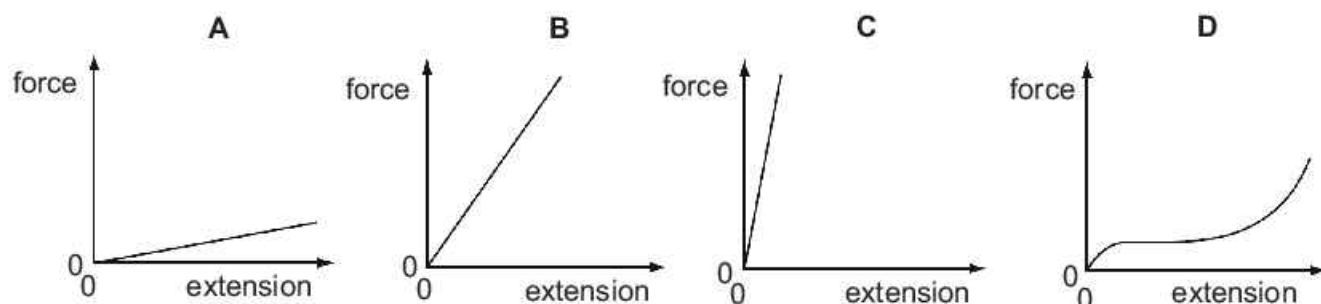
How much work needs to be done by the tensile force, to two significant figures, to cause an extension of 7.0 mm?

- A** 0.088 J **B** 0.12 J
C 0.53 J **D** 120 J

66 9702/12/O/N/11/Q23

The following force-extension graphs are drawn to the same scale.

Which graph represents the deformed object with the greatest amount of elastic potential energy?



67 9702/12/M/J/12/Q25

A wire stretches 8 mm under a load of 60 N.

A second wire of the same material, with half the diameter and a quarter of the original length of the first wire, is stretched by the same load.

Assuming that Hooke's law is obeyed, what is the extension of this wire?

- A** 1 mm **B** 4 mm **C** 8 mm **D** 16 mm

68 9702/13/M/J/12/Q20

What is represented by the gradient of a graph of force (vertical axis) against extension (horizontal axis)?

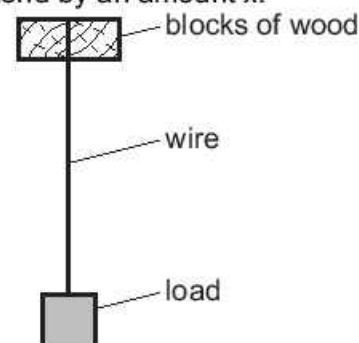
- A** elastic limit **B** spring constant **C** stress **D** Young modulus

69 9702/13/M/J/12/Q21

The diagram shows a wire of diameter D and length L that is firmly clamped at one end between two blocks of wood. A load is applied to the wire which causes it to extend by an amount x .

By how much would a wire of the same material, but of diameter $2D$ and length $3L$, extend when the same load is applied?

- A** $\frac{2}{3}x$ **B** $\frac{3}{4}x$
C $\frac{4}{3}x$ **D** $\frac{3}{2}x$



70 9702/13/M/J/12/Q22

Which property of a metal wire depends on its Young modulus?

- A** ductility **B** elastic limit **C** spring constant **D** ultimate tensile stress

71 9702/13/M/J/12/Q23

When white sugar granules are heated, they melt. When the melt is cooled quickly, a brittle solid form of toffee is produced.

How does the structure of the sugar change?

- A** amorphous to polymeric **C** crystalline to polymeric
B crystalline to amorphous **D** polymeric to amorphous

72 9702/12/O/N/12/Q24

Which row gives the correct description for the arrangement of atoms in the four types of material?

	atoms have no long-range order	atoms form giant chain-like molecules	atoms have an ordered arrangement in regions, but these ordered regions are at angles to one another	atoms are arranged in an ordered way throughout
A	crystalline	amorphous	polymeric	polycrystalline
B	polycrystalline	crystalline	amorphous	polymeric
C	polymeric	polycrystalline	crystalline	amorphous
D	amorphous	polymeric	polycrystalline	crystalline

73 9702/12/O/N/12/Q25

What leads to the conclusion that the movement of molecules is random?

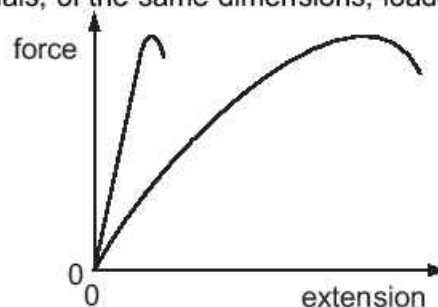
- A evaporation of water at room temperature
- B conduction of electricity in water
- C convection currents in air
- D motion of dust particles in air

74 9702/12/O/N/12/Q26

The diagram shows the force-extension graphs for two materials, of the same dimensions, loaded to fracture.

What describes the behaviour of the materials?

- A Both materials are brittle.
- B Both materials obey Hooke's law.
- C Both materials are plastic.
- D Both materials have the same ultimate tensile stress.



75 9702/12/O/N/12/Q27

Two wires, X and Y, are made from different metals and have different dimensions. The Young modulus of wire X is twice that of wire Y. The diameter of wire X is half that of wire Y.

Both wires are extended with equal strain and obey Hooke's law.

What is the ratio $\frac{\text{tension in wire X}}{\text{tension in wire Y}}$?

- A $\frac{1}{8}$
- B $\frac{1}{2}$
- C 1
- D 8

76 9702/13/O/N/12/Q24

A trolley is held at rest between two steel springs.

Each spring has an unstretched length of 0.10 m.

Spring P has spring constant 60 N m^{-1} .

Spring Q has spring constant 120 N m^{-1} .

Spring P has an extension of 0.40 m.



What is the extension of spring Q?

- A 0.10 m
- B 0.20 m
- C 0.30 m
- D 0.80 m

77 9702/13/O/N/12/Q25

A lift is supported by two steel cables, each of length 10 m and diameter 0.5 cm.

The lift drops 1 mm when a man of mass 80 kg steps into the lift.

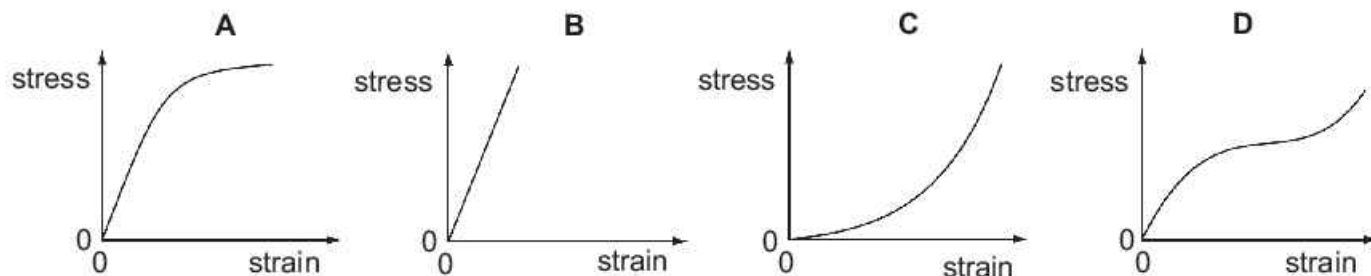
What is the best estimate of the value of the Young modulus of the steel?

- A** $2 \times 10^{10} \text{ Nm}^{-2}$ **B** $4 \times 10^{10} \text{ Nm}^{-2}$ **C** $2 \times 10^{11} \text{ Nm}^{-2}$ **D** $4 \times 10^{11} \text{ Nm}^{-2}$

78 9702/11/M/J/13/Q22

The stress-strain graphs for four different materials are shown below.

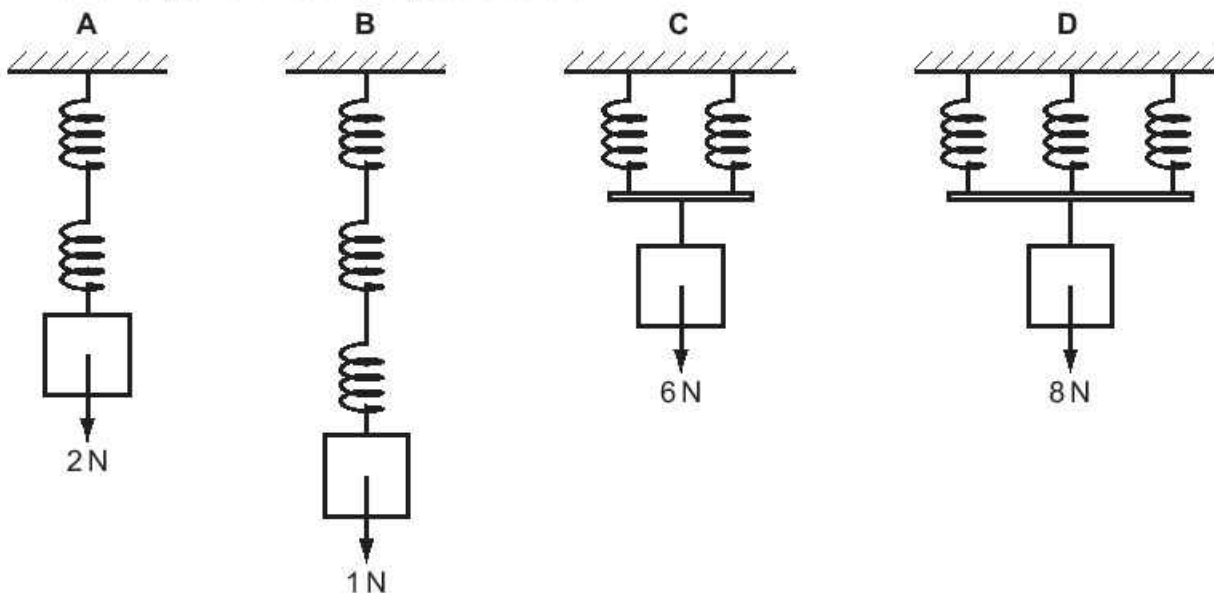
Which diagram shows the stress-strain graph for a ductile metal?



79 9702/11/M/J/13/Q23

A number of identical springs, each having the same spring constant, are joined in four arrangements. A different load is applied to each arrangement.

Which arrangement has the largest extension?



80 9702/01/O/N/06/Q23

What is the unit of the Young modulus?

- A** Nm^{-1} **B** Nm **C** Nm^{-2} **D** Nm^2

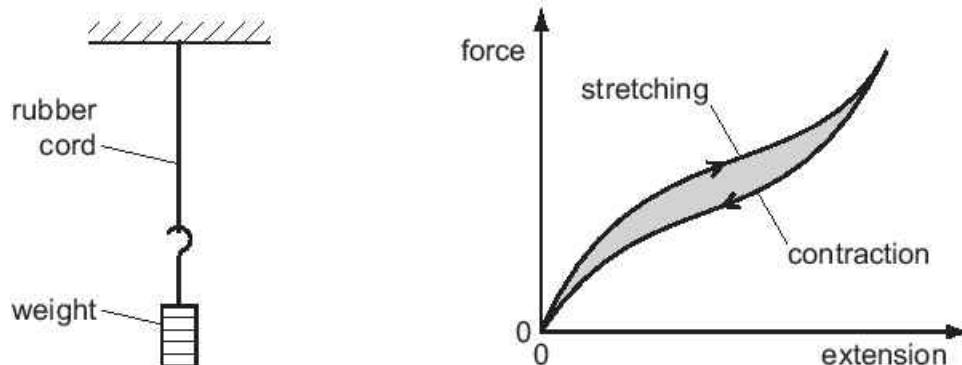
81 9702/12/O/N/13/Q21

Which two substances are normally both crystalline?

- A** copper and diamond **C** diamond and glass
B copper and glass **D** diamond and rubber

82 9702/12/M/J/13/Q22

A rubber cord hangs from a rigid support. A weight attached to its lower end is gradually increased from zero, and then gradually reduced to zero.



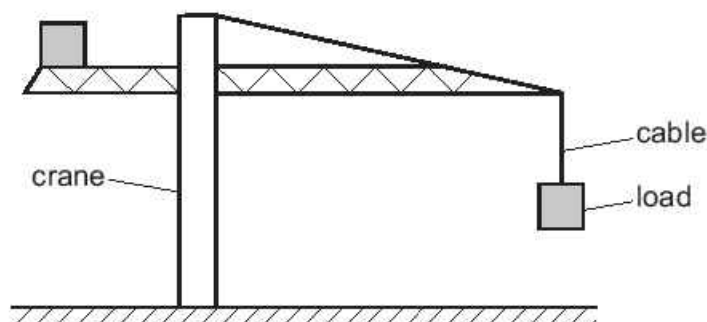
The force-extension curve for contraction is below the force-extension curve for stretching.

What does the shaded area between the curves represent?

- A the amount of elastic energy stored in the rubber
- B the amount of thermal energy dissipated in the rubber
- C the work done on the rubber cord during stretching
- D the work done by the rubber cord during contraction

83 9702/12/M/J/13/Q23

The diagram shows a large crane on a construction site lifting a cube-shaped load.



A model is made of the crane, its load and the cable supporting the load.

The material used for each part of the model is the same as that in the full-size crane, cable and load. The model is one tenth full-size in all linear dimensions.

What is the ratio $\frac{\text{stress in the cable on the full-size crane}}{\text{stress in the cable on the model crane}}$?

- A 10^0
- B 10^1
- C 10^2
- D 10^3

84 9702/13/O/N/13/Q22

A lift is supported by two steel cables each of length 20m.

Each of the cables consists of 100 parallel steel wires, each wire of cross-sectional area $3.2 \times 10^{-6} \text{ m}^2$. The Young modulus of steel is $2.1 \times 10^{11} \text{ N m}^{-2}$.

Which distance does the lift move downward when a man of mass 70kg steps into it?

- A 0.010mm
- B 0.020mm
- C 0.10mm
- D 0.20mm

85 9702/01/O/N/08/Q19

Which properties best describe modelling clay?

- A brittle and ductile
- B ductile and elastic
- C elastic and plastic
- D plastic and ductile

86 9702/12/O/N/13/Q24

A steel spring has a spring constant of 150 N m^{-1} . When a 25 N weight is hung from the spring, it has a stretched length of 55 cm.

What was the original length of the spring?

- A 0.38 m
- B 0.49 m
- C 0.61 m
- D 0.72 m

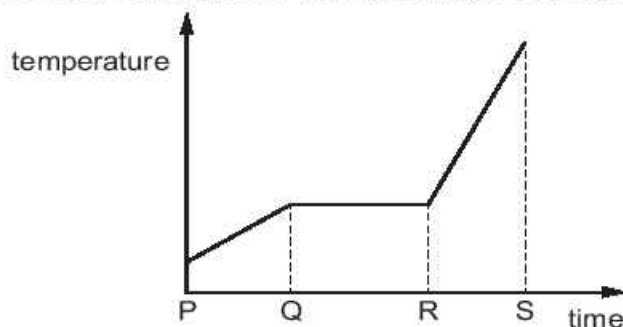
87 9702/13/O/N/13/Q23

What is equal to the Young modulus of a material that is extended elastically within the limit of proportionality?

- A area under the force-extension graph
- B area under the stress-strain graph
- C gradient of the force-extension graph
- D gradient of the stress-strain graph

88 9702/11/M/J/14/Q19

A crystalline solid is heated at a constant rate and the change of temperature with time is shown in the graph below.



Which statement about the particles in the material is correct?

- A In the time from P to Q, the particles are arranged randomly.
- B In the time from Q to R, some particles are arranged regularly and some particles are arranged randomly.
- C In the time from R to S, the particles are widely spaced.
- D The arrangement of the particles is the same in the time from P to S.

89 9702/13/M/J/14/Q18

A ball drops onto a horizontal surface and bounces elastically.

What happens to the kinetic energy of the ball during the very short time that it is in contact with the surface?

- A Most of the kinetic energy is lost as heat and sound energy.
- B The kinetic energy decreases to zero and then returns to its original value.
- C The kinetic energy remains constant because it is an elastic collision.
- D The kinetic energy remains constant in magnitude but changes direction.

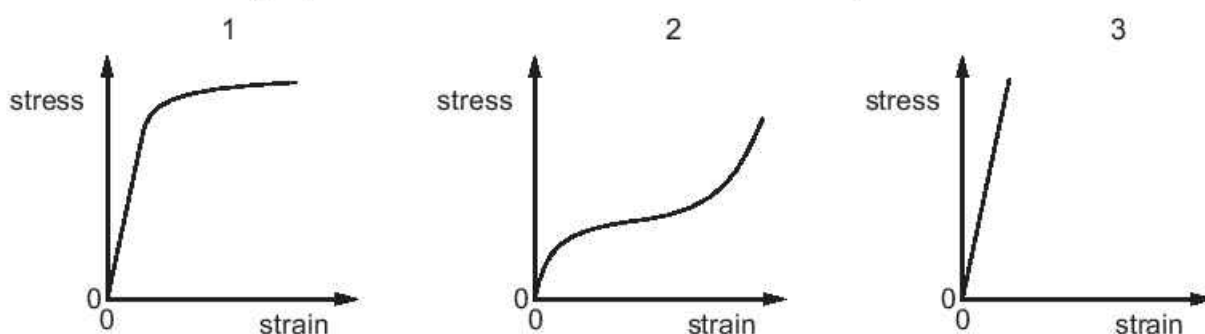
90 9702/13/M/J/14/Q20

Which process does **not** require energy to be supplied?

- A boiling
- B evaporation
- C freezing
- D melting

91 9702/11/M/J/14/Q20

The stress-strain graphs for three different materials are shown, not drawn to the same scales.



The three materials are copper, rubber and glass.
Which materials are represented by the graphs?

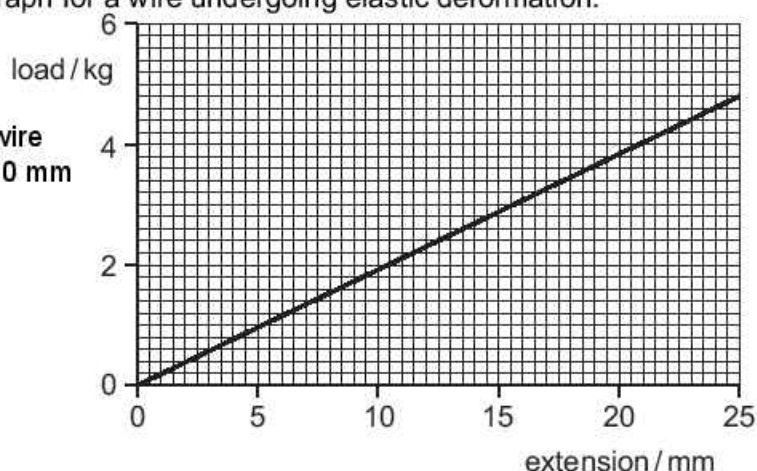
	1	2	3
A	copper	glass	rubber
B	copper	rubber	glass
C	glass	copper	rubber
D	glass	rubber	copper

92 9702/11/M/J/14/Q21

The graph is a load-extension graph for a wire undergoing elastic deformation.

How much work is done on the wire to increase the extension from 10 mm to 20 mm?

- A 0.028 J B 0.184 J
C 0.28 J D 0.37 J

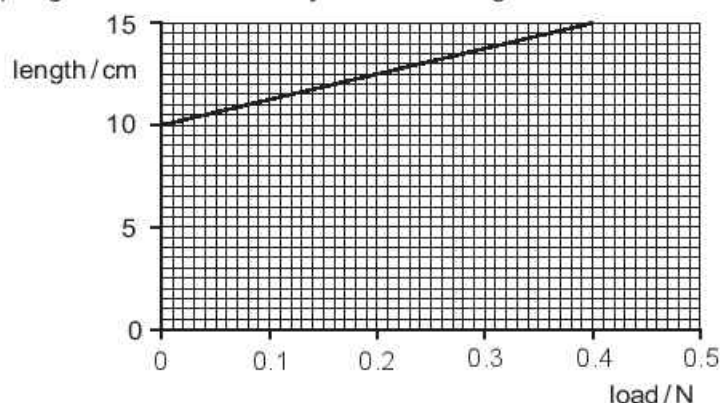


93 9702/12/M/J/14/Q20

The graph shows the length of a spring as it is stretched by an increasing load.

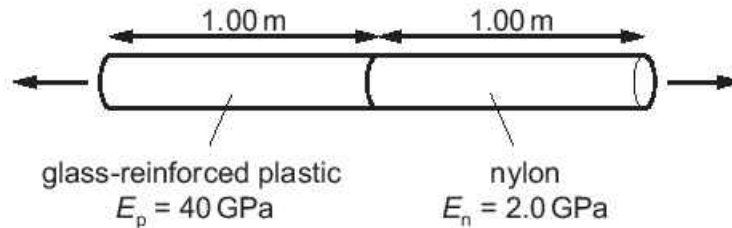
What is the spring constant?

- A 8.0 N m^{-1}
B 2.7 N m^{-1}
C 0.13 N m^{-1}
D 0.080 N m^{-1}



94 9702/12/M/J/14/Q21

A composite rod is made by attaching a glass-reinforced plastic rod and a nylon rod end to end, as shown.



The rods have the same cross-sectional area and each rod is 1.00 m in length. The Young modulus E_p of the plastic is 40 GPa and the Young modulus E_n of the nylon is 2.0 GPa.

The composite rod will break when its total extension reaches 3.0 mm.

What is the greatest tensile stress that can be applied to the composite rod before it breaks?

- A $7.1 \times 10^{-14} \text{ Pa}$ B $7.1 \times 10^{-2} \text{ Pa}$ C $5.7 \times 10^6 \text{ Pa}$ D $5.7 \times 10^9 \text{ Pa}$

95 9702/13/M/J/14/Q19

When the water in a pond freezes, it changes from a liquid to a solid. When this occurs, it changes volume and exchanges energy with the surroundings.

Which row is correct?

	change of volume	energy exchange
A	contracts	gives out energy to the surroundings
B	contracts	takes in energy from the surroundings
C	expands	gives out energy to the surroundings
D	expands	takes in energy from the surroundings

96 9702/13/M/J/14/Q23

An elastic material with a Young modulus E is subjected to a tensile stress S . Hooke's Law is obeyed.

What is the expression for the elastic energy stored per unit volume of the material?

- A $\frac{S^2}{2E}$ B $\frac{S^2}{E}$ C $\frac{E}{2S^2}$ D $\frac{2E}{S^2}$

97 9702/12/O/N/14/Q18

Which statement about molecules in a gas is correct?

- A In Brownian motion experiments, the molecules can be seen moving randomly in all directions.
 B The pressure exerted by a gas is caused by molecules bouncing against each other and changing kinetic energy.
 C The pressure exerted by a gas is caused by molecules rebounding from the walls of a container and changing momentum.
 D When the average speed of the molecules in a closed container increases, the density must also increase.

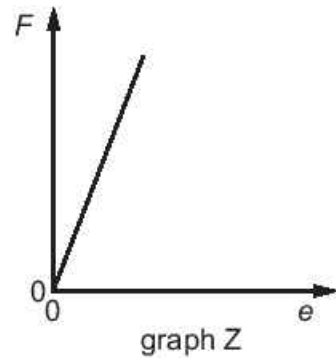
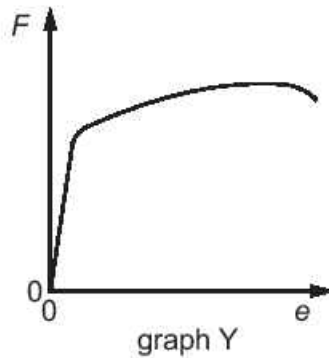
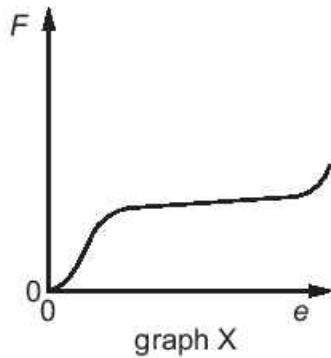
98 9702/12/O/N/14/Q17

What is the correct name for a material containing long-chain molecules that are tangled and coiled?

- A amorphous metal C crystalline metal
 B amorphous polymer D crystalline polymer

99 9702/13/M/J/14/Q24

Cylindrical samples of steel, glass and rubber are each subjected to a gradually increasing tensile force F . The extensions e are measured and graphs are plotted as shown below.



Which row correctly relates the graphs to the materials?

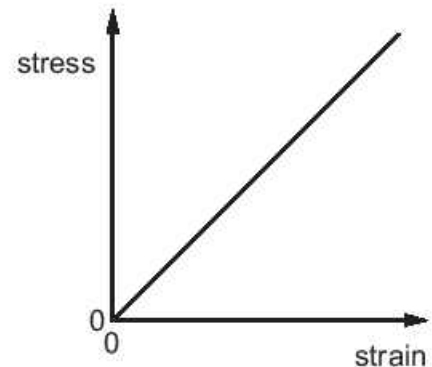
	steel	glass	rubber
A	X	Y	Z
B	X	Z	Y
C	Y	X	Z
D	Y	Z	X

100 9702/11/O/N/14/Q20

The stress-strain graph for a glass rod, up to the point at which it breaks, is shown below.

Which statement about the glass rod is correct?

- A Hooke's law is obeyed for all values of stress up to the breaking point.
- B The glass is ductile.
- C The glass shows plastic deformation.
- D When the cross-sectional area of the rod is doubled, the ultimate tensile stress of the rod is halved.

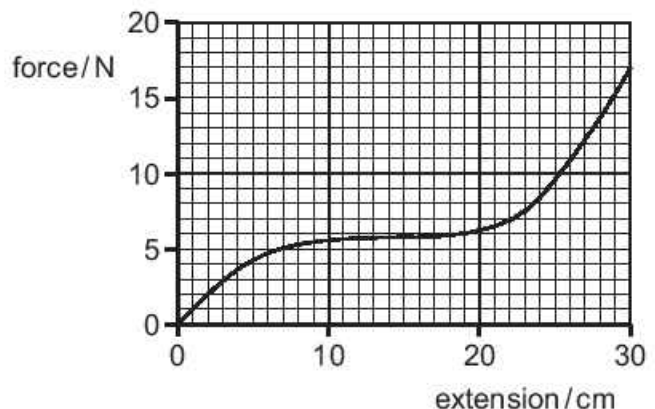


101 9702/11/O/N/14/Q21

A rubber band is stretched by hanging weights on it and the force-extension graph is plotted from the results.

What is the best estimate of the strain energy stored in the rubber band when it is extended 30 cm?

- A 1.8 J
- B 2.6 J
- C 5.1 J
- D 200 J



102 9702/13/O/N/14/Q21

A student is studying Brownian motion.

Using a microscope, she observes particles of smoke in a glass container, illuminated by a strong light. The particles of smoke have a zig-zag path, constantly changing speed and direction.

What happens to the smoke particles if the air in the container is heated?

- A The smoke particles become easier to see.
- B The smoke particles change direction more frequently.
- C The smoke particles increase in volume.
- D The smoke particles move further apart.

103 9702/13/O/N/14/Q23

What is meant by the ultimate tensile stress of a material?

- A the maximum force that can be applied to a bar of the material before it bends
- B the maximum inter-atomic force before the atomic bonds of the material break
- C the maximum stretching force per unit cross-sectional area before the material breaks
- D the maximum tensile force in a wire of the material before it breaks

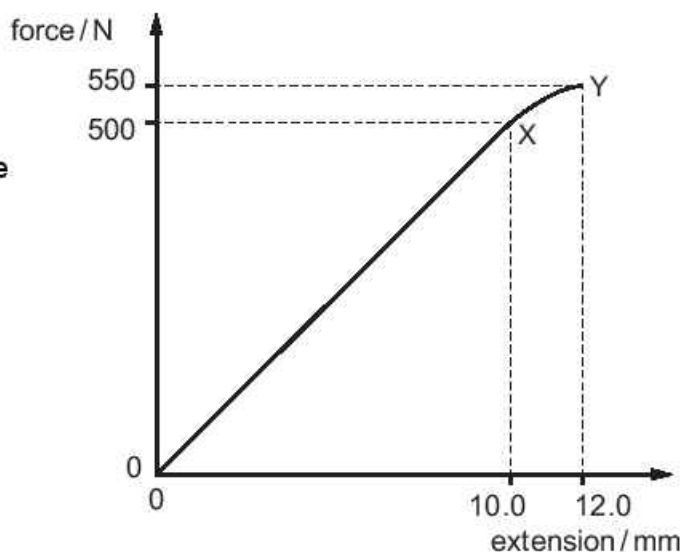
104 9702/13/O/N/14/Q24

The graph shows the behaviour of a sample of a metal when it is stretched until it starts to undergo plastic deformation.

What is the total work done in stretching the sample from zero to 12.0 mm extension?

Simplify the calculation by treating the curve XY as a straight line.

- A 3.30 J B 3.55 J
- C 3.60 J D 6.60 J

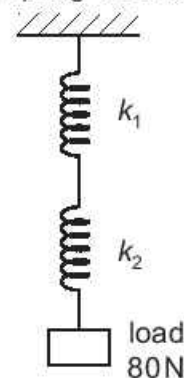


105 9702/13/O/N/14/Q25

Two springs, one with spring constant $k_1 = 4 \text{ kN m}^{-1}$ and the other with spring constant $k_2 = 2 \text{ kN m}^{-1}$, are connected as shown.

What is the total extension of the springs when supporting a load of 80 N?

- A 1.3 cm B 4 cm C 6 cm D 60 cm



106 9702/11/M/J/15/Q20

Descriptions of three different types of material are listed.

- 1 a polycrystalline material made up of large numbers of small crystals
- 2 an amorphous material with little or no ordered arrangement of molecules
- 3 a polymeric material consisting of long chains of molecules

Which row correctly matches the descriptions to nylon, copper and glass?

	1	2	3
A	copper	glass	nylon
B	copper	nylon	glass
C	glass	nylon	copper
D	nylon	copper	glass

107 9702/11/M/J/15/Q21

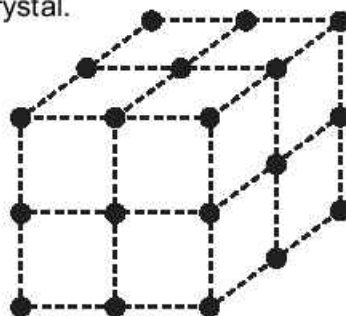
The diagram shows the arrangement of atoms in a particular crystal.

The mass of each atom is 3.5×10^{-25} kg.

The density of the crystal is 9.2×10^3 kg m⁻³.

What is the shortest distance between the centres of two adjacent atoms?

- A 3.8×10^{-29} m B 6.2×10^{-15} m
C 3.4×10^{-10} m D 3.0×10^{-9} m



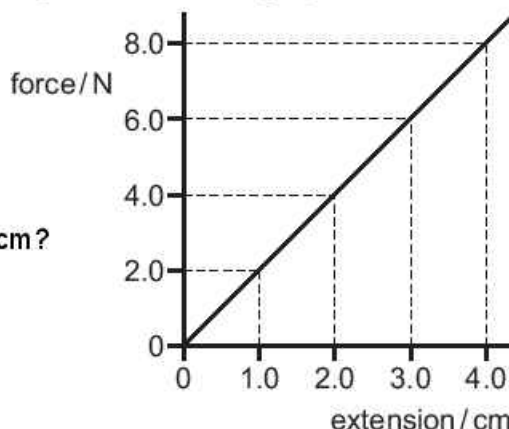
108 9702/11/M/J/15/Q23

The variation with applied force of the extension of a spring is shown in the graph.

When there is no force applied to the spring, it has a length of 1.0 cm.

What is the increase in the strain energy stored in the spring when its length is increased from 2.0 cm to 3.0 cm?

- A 0.020 J B 0.030 J
C 0.040 J D 0.050 J



109 9702/12/M/J/15/Q21

To determine the Young modulus of a wire, several measurements are taken.

In which row can the measurement **not** be taken **directly** with the stated apparatus?

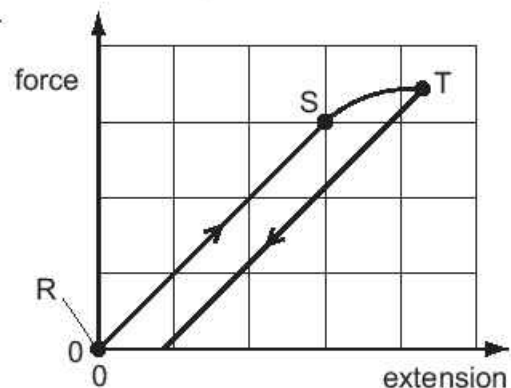
	measurement	apparatus
A	area of cross-section of wire	micrometer screw gauge
B	extension of wire	vernier scale
C	mass of load applied to wire	electronic balance
D	original length of wire	metre rule

110 9702/12/M/J/15/Q22

A long, thin metal wire is suspended from a fixed support and hangs vertically. Masses are suspended from its lower end.

The load on the lower end is increased from zero and then decreased again back to zero.

The diagram shows the force-extension graph produced.

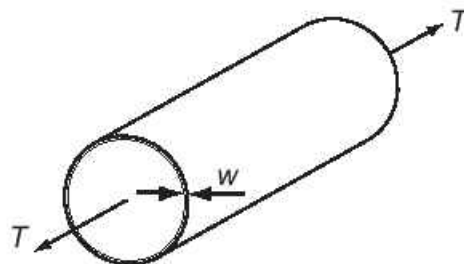


Where on the graph would the elastic limit be found?

- A anywhere between point R and point S
- B just beyond point S
- C exactly at point S
- D exactly at point T

111 9702/12/M/J/15/Q23

The diagram represents a steel tube with wall thickness w which is small in comparison with the diameter of the tube.



The tube is under tension, caused by a force T , parallel to the axis of the tube. To reduce the stress in the material of the tube, it is proposed to thicken the wall.

The tube diameter and the tension being constant, which wall thickness gives half the stress?

- A $\frac{w}{2}$
- B $\sqrt{2}w$
- C $2w$
- D $4w$

112 9702/13/M/J/15/Q20

When ice melts, it contracts.

Which row is correct for ice turning into water?

	distance between molecules	density
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

113 9702/13/M/J/15/Q22

A steel bar of circular cross-section is under tension T , as shown.

The diameter of the wide portion is double the diameter of the narrow portion.



What is the value of $\frac{\text{stress in the wide portion}}{\text{stress in the narrow portion}}$?

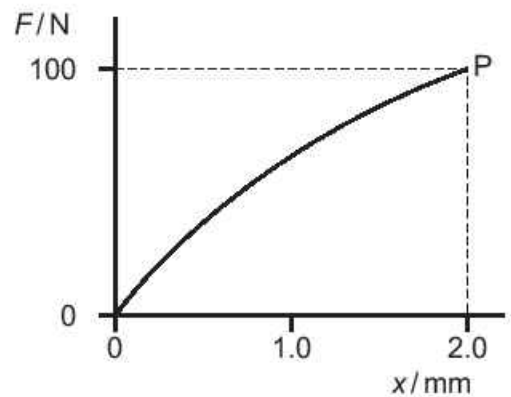
- A 0.25 B 0.50 C 2.0 D 4.0

114 9702/13/M/J/15/Q23

The graph shows the non-linear force-extension curve for a wire made from a new composite material.

What could be the value of the strain energy stored in the wire when it is stretched elastically to point P?

- A 0.09 J B 0.10 J
C 0.11 J D 0.20 J



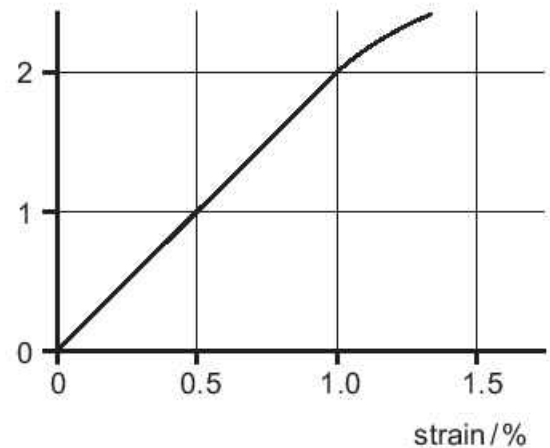
115 9702/13/M/J/15/Q24

The diagram shows the stress-strain graph for bone.

What is the Young modulus of bone?

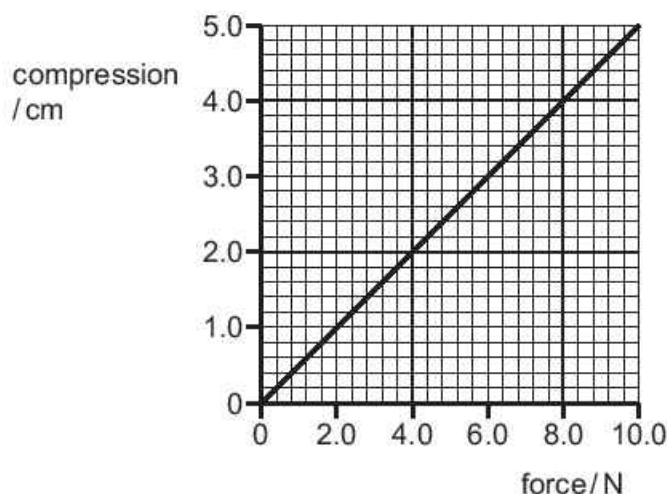
- A $1 \times 10^6 \text{ Nm}^{-2}$ B $2 \times 10^6 \text{ Nm}^{-2}$
C $1 \times 10^8 \text{ Nm}^{-2}$ D $2 \times 10^8 \text{ Nm}^{-2}$

stress
 $/ 10^6 \text{ Nm}^{-2}$

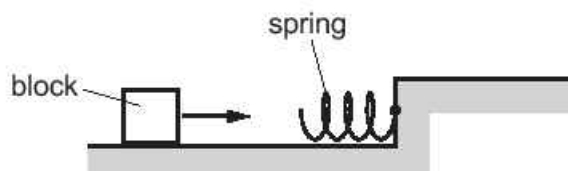


116 9702/11/M/J/17/Q21

The variation of the compression of a spring with the force applied to it is shown in the graph.



A block slides along a horizontal frictionless surface towards the spring, as shown.



The block is brought to rest by the spring. When the spring reaches a compression of 4.0 cm, all of the kinetic energy of the block is transferred to the elastic potential energy of the spring.

What is the kinetic energy of the block when it first makes contact with the spring?

- A 0.16 J B 0.32 J C 16 J D 32 J

117 9702/11/M/J/17/Q20

A wire of diameter d and length l hangs vertically from a fixed point. The wire is extended by hanging a mass M on its end. The Young modulus of the wire is E . The acceleration of free fall is g .

Which equation is used to determine the extension x of the wire?

- A $x = \frac{Ml}{\pi d^2 E}$ B $x = \frac{Mgl}{\pi d^2 E}$ C $x = \frac{4Mgl}{\pi d E}$ D $x = \frac{4Mgl}{\pi d^2 E}$

118 9702/12/M/J/17/Q20

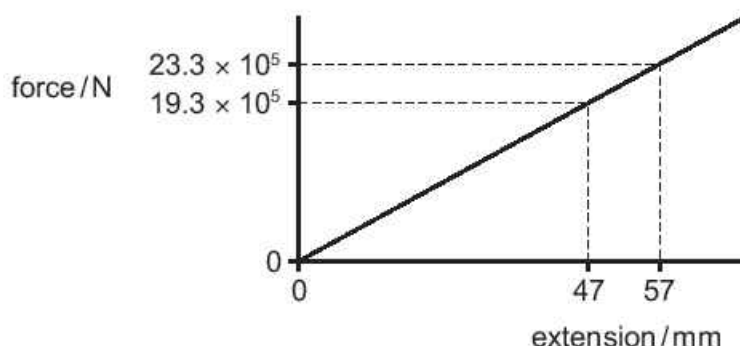
What are the units of stress, strain and the Young modulus?

	stress	strain	Young modulus
A	newton	metre	pascal
B	newton	no unit	newton
C	pascal	metre	newton
D	pascal	no unit	pascal

119 9702/12/M/J/17/Q19

A cable on a suspension bridge supports a weight of $19.3 \times 10^5 \text{ N}$. This weight causes the cable to stretch by 47 mm.

A lorry crossing the bridge then increases the force on the cable to $23.3 \times 10^5 \text{ N}$. The force-extension graph for the cable is shown.

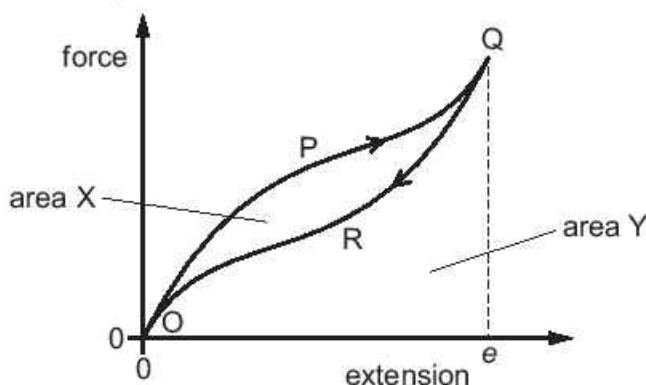


What is the **increase** in strain energy in the cable when the lorry is crossing the bridge?

- A 21 kJ B 23 kJ C 45 kJ D 66 kJ

120 9702/12/M/J/17/Q21

A rubber band is stretched and then relaxed to its original length. The diagram shows the force-extension graph for this process.



As the force is increased, the curve follows the path OPQ to extension e . As the force is reduced, the curve follows the path QRO to return to zero extension.

The area labelled X is between the curves OPQ and QRO. The area labelled Y is bounded by the curve QRO and the horizontal axis.

Which statement about the process is correct?

- A Area X is the energy which heats the band as it is stretched to extension e .
 B (Area X + area Y) is the minimum energy required to stretch the band to extension e .
 C Area X is the elastic potential energy stored in the band when it is stretched to extension e .
 D (Area Y – area X) is the net work done on the band during the process.

121 9702/11/O/N/17/Q20

A spring is loaded with weights. When the weights are removed, the spring returns to its original length.

The spring is then loaded with heavier weights. When the weights are removed, the spring is longer than it was originally.

Which types of deformation are shown by this experiment?

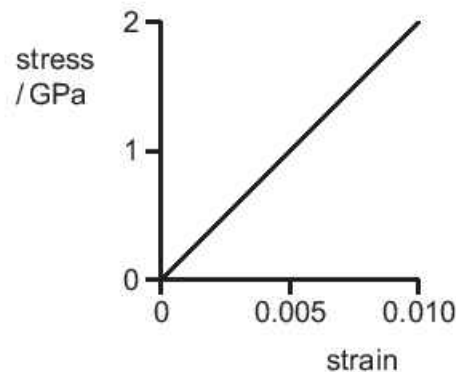
- A** both elastic and plastic deformation **C** neither elastic nor plastic deformation
B elastic deformation only **D** plastic deformation only

122 9702/11/O/N/17/Q21

The stress-strain graph for a metal is shown.

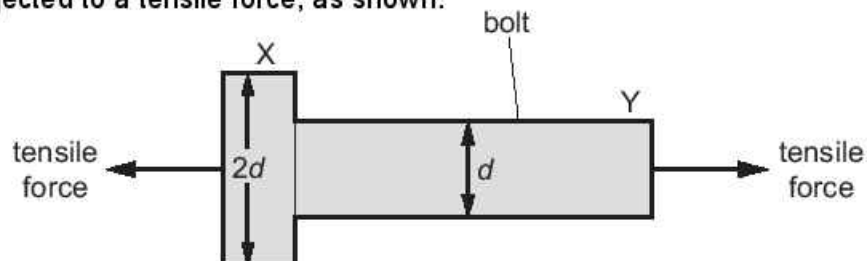
What is the strain energy per unit volume of a rod made from this metal when the strain of the rod is 0.010?

- A** 10 kJ m^{-3}
B 100 kJ m^{-3}
C 1.0 MJ m^{-3}
D 10 MJ m^{-3}



123 9702/12/O/N/17/Q20

A bolt is subjected to a tensile force, as shown.



The bolt has a circular cross-section. At end X the diameter is $2d$. At end Y the diameter is d .

What is the ratio $\frac{\text{stress at Y}}{\text{stress at X}}$?

- A** 0.25 **B** 0.50 **C** 2.0 **D** 4.0

124 9702/12/O/N/17/Q21

A rectangular block of steel supporting a very large component of a bridge has a height of 15 cm and a cross-section of $20 \text{ cm} \times 12 \text{ cm}$. It is designed to compress 1 mm when under maximum, evenly distributed, load.

The Young modulus of steel is $2.0 \times 10^{11} \text{ N m}^{-2}$.

What is the maximum load it can support?

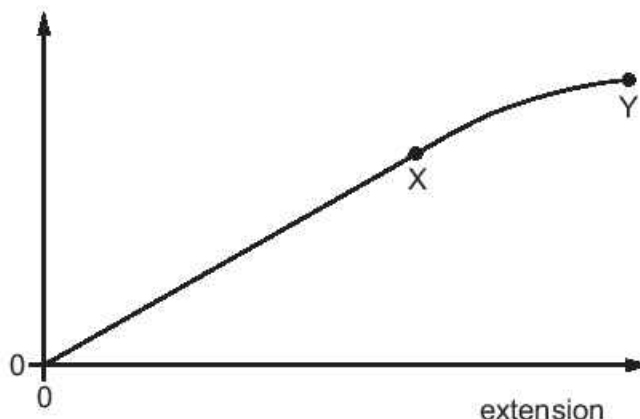
- A** 32 MN **B** 56 GN **C** 720 GN **D** 32 TN

125 9702/13/O/N/17/Q20

A sample of metal is subjected to a force which increases to a maximum value and then decreases back to zero. A force-extension graph for the sample is shown.

When the sample contracts, it follows the same force-extension curve as when it was being stretched. What is the behaviour of the metal between X and Y?

- A both elastic and plastic
- B not elastic and not plastic
- C plastic but not elastic
- D elastic but not plastic



126 9702/13/O/N/17/Q21

A load is hung from the end of a metal wire. The load is increased and the wire stretches elastically. The table shows the length of the wire for different loads.

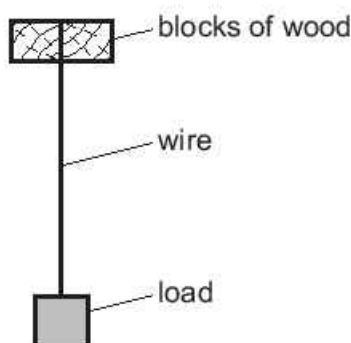
load / kN	length / mm
0	500.0
1.0	502.0
2.0	504.0
3.0	506.0
4.0	508.0

When the load is 4.0 kN, what is the strain energy stored in the wire?

- A 16 J
- B 32 J
- C 1.0 kJ
- D 2.0 kJ

127 9702/11/M/J/18/Q18

The diagram shows a wire of diameter D and length L that is firmly clamped at one end between two blocks of wood. A load is applied to the wire which extends its length by x .



A second wire is made of the same material, but of diameter $2D$ and length $3L$. Both wires obey Hooke's law.

What is the extension of the second wire when the same load is applied?

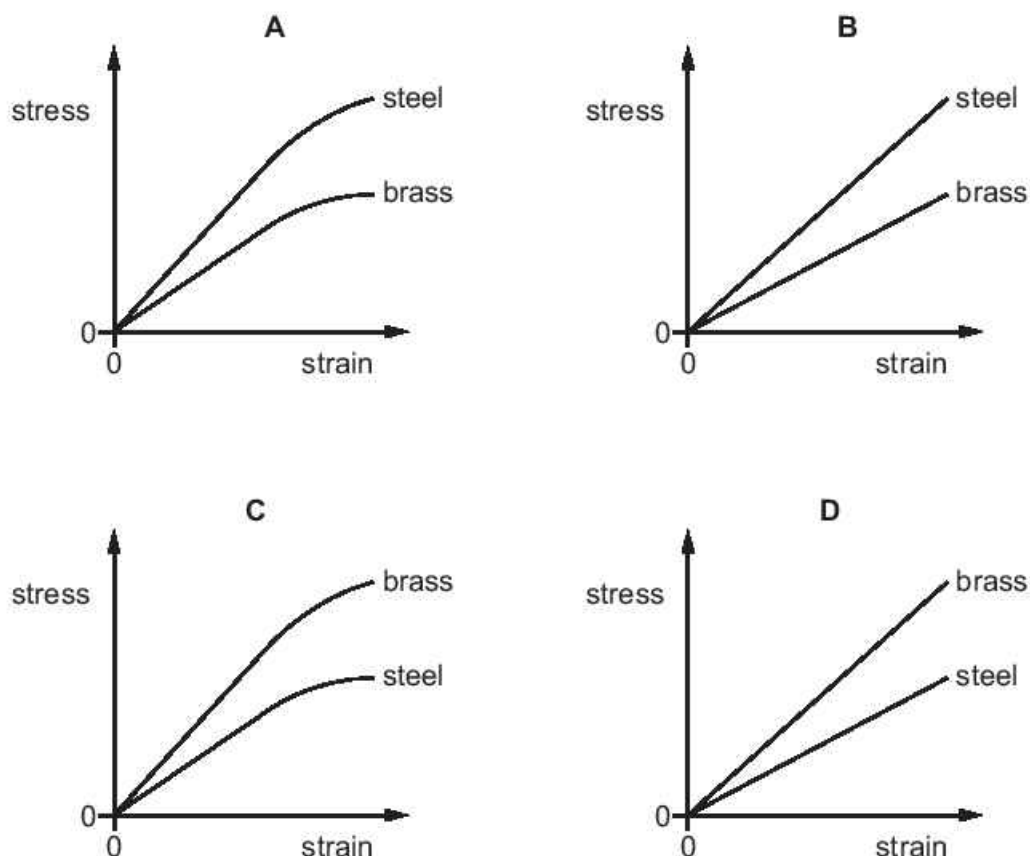
- A $\frac{2}{3}x$
- B $\frac{3}{4}x$
- C $\frac{4}{3}x$
- D $\frac{3}{2}x$

128 9702/11/M/J/18/Q19

Two wires, one made of brass and the other of steel, are stretched in an experiment. Both wires obey Hooke's law during this experiment.

The Young modulus for brass is less than the Young modulus for steel.

Which graph shows how the stress varies with strain for both wires in this experiment?



129 9702/12/M/J/18/Q20

An elastic material with Young modulus E is subjected to a tensile stress S . Hooke's law is obeyed.

What is the expression for the elastic energy stored per unit volume of the material?

A $\frac{E}{2S^2}$

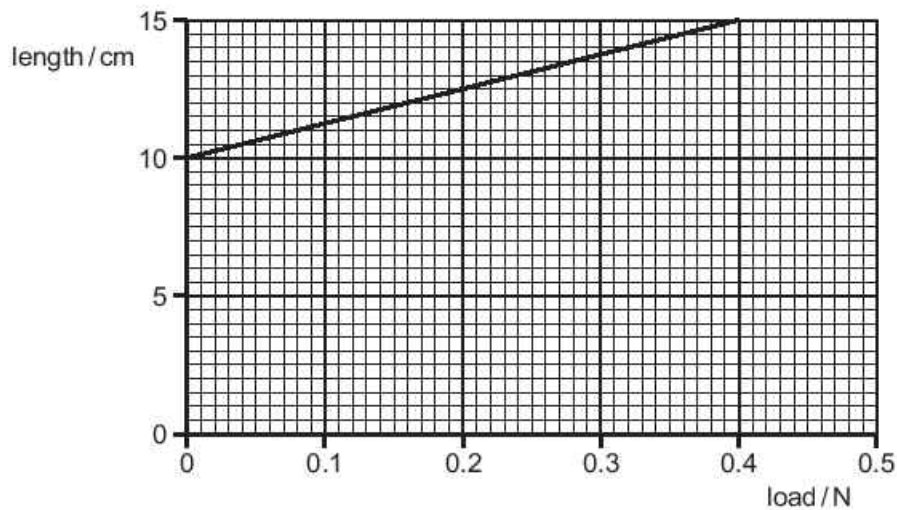
B $\frac{2E}{S^2}$

C $\frac{S^2}{E}$

D $\frac{S^2}{2E}$

130 9702/12/M/J/18/Q21

The graph shows the length of a spring as it is stretched by an increasing load.



What is the spring constant of the spring?

- A** 0.080 Nm^{-1} **B** 0.13 Nm^{-1} **C** 2.7 Nm^{-1} **D** 8.0 Nm^{-1}

131 9702/13/M/J/18/Q18

Data for a steel wire on an electric guitar are listed.

diameter = $5.0 \times 10^{-4} \text{ m}$

Young modulus = $2.0 \times 10^{11} \text{ Pa}$

tension = 20 N

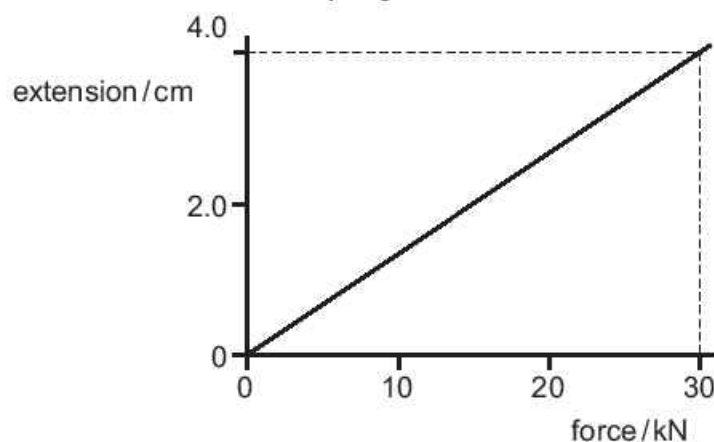
The wire snaps and contracts elastically. Assume the wire obeys Hooke's law.

By what percentage does the length l of a piece of the wire contract?

- A** $1.3 \times 10^{-4} \%$ **B** $5.1 \times 10^{-4} \%$ **C** $1.3 \times 10^{-2} \%$ **D** $5.1 \times 10^{-2} \%$

132 9702/13/M/J/18/Q19

The graph shows how the extension of a spring varies with the force used to stretch it.



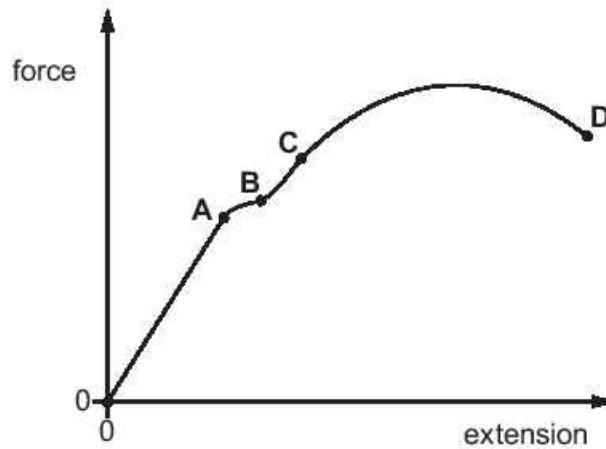
What is the strain energy in the spring when the extension is 4.0 cm ?

- A** 60 J **B** 120 J **C** 600 J **D** 1200 J

133 9702/12/F/M/18/Q19

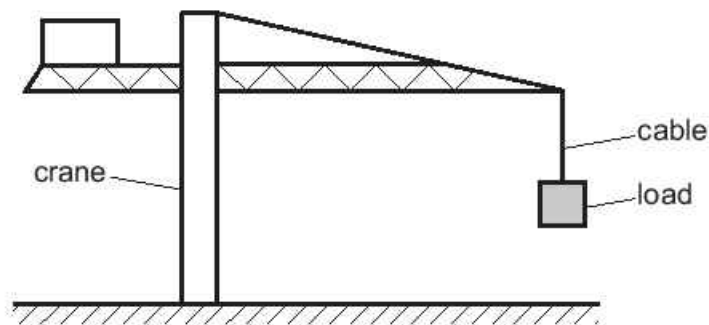
The force-extension graph of a metal wire is shown.

At which point on the graph does the metal wire stop obeying Hooke's law?



134 9702/12/F/M/18/Q20

The diagram shows a large crane on a construction site lifting a cube-shaped load at a constant speed.



A model is made of the crane, its load and the cable supporting the load.

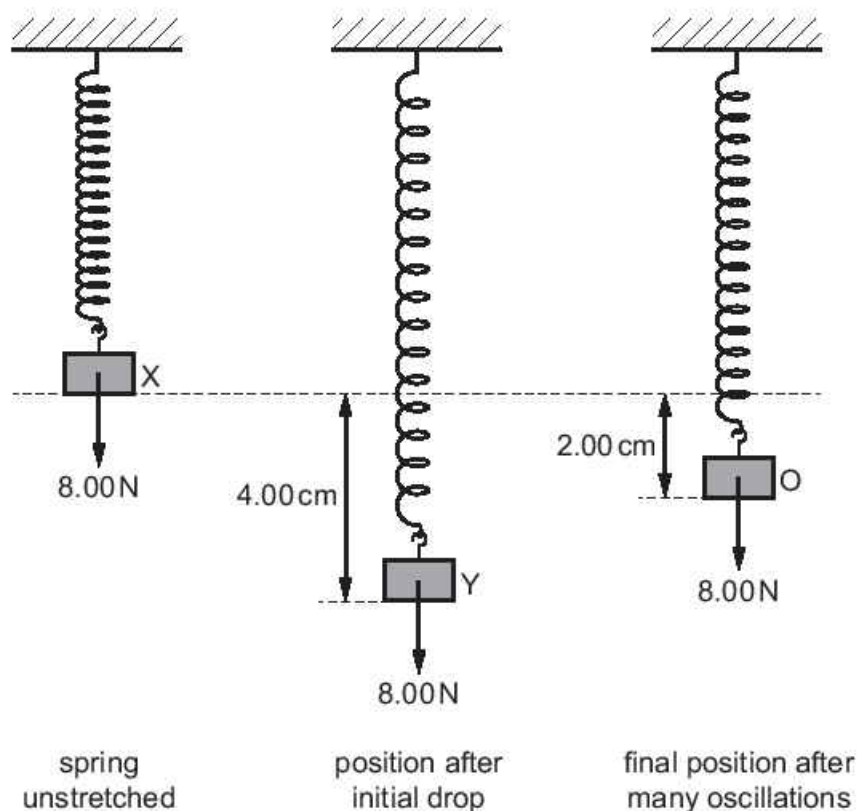
The material used for each part of the model is the same as that in the full-size crane, cable and load. The model is one tenth full-size in all linear dimensions.

What is the ratio $\frac{\text{stress in the cable on the full-size crane}}{\text{stress in the cable on the model crane}}$?

- A** 0.1 **B** 1 **C** 10 **D** 100

135 9702/11/O/N/18/Q16

An 8.00 N weight is attached to the lower end of a spring which is fixed at its upper end. The weight is initially held at rest at position X and the spring is unstretched. The weight is then released and falls to position Y, which is 4.00 cm below X. The weight oscillates and then eventually comes to rest at O, which is 2.00 cm below X.

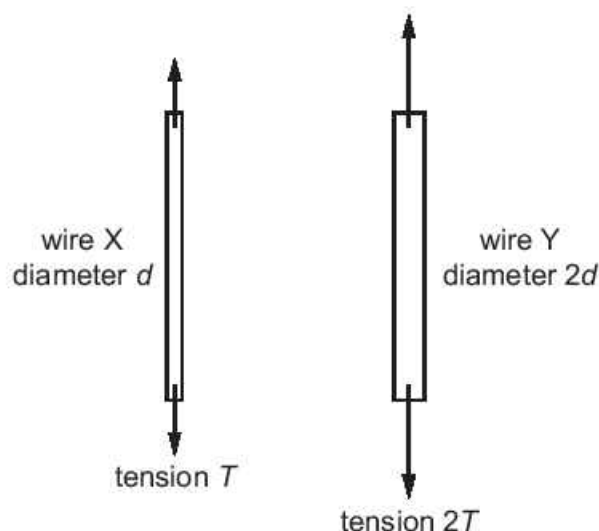


How much energy is lost from the system?

- A** 0.04 J **B** 0.08 J **C** 0.16 J **D** 0.32 J

136 9702/11/O/N/18/Q18

Two wires X and Y are made from the same material. Wire Y has twice the diameter and experiences twice the tension of wire X. The wires obey Hooke's law and have the same original length.



Wire X has extension e .

What is the extension of wire Y?

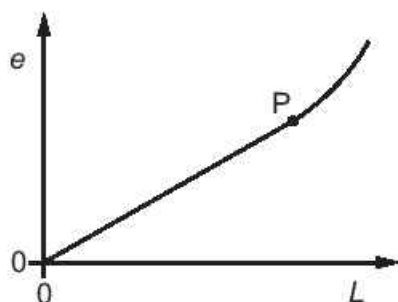
- A $\frac{e}{4}$ B $\frac{e}{2}$ C e D $2e$

137 9702/11/O/N/18/Q19

What is represented by the gradient of a graph of force (vertical axis) against extension (horizontal axis) for a wire obeying Hooke's law?

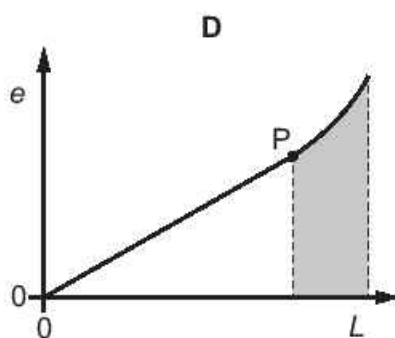
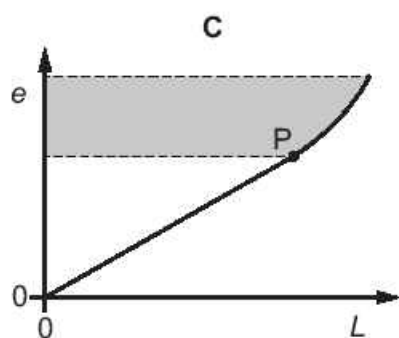
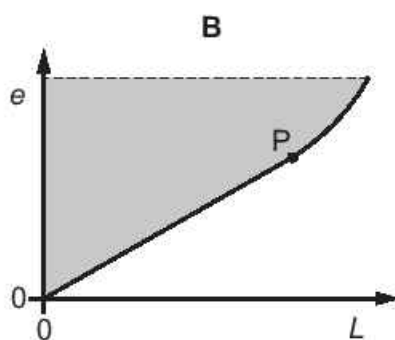
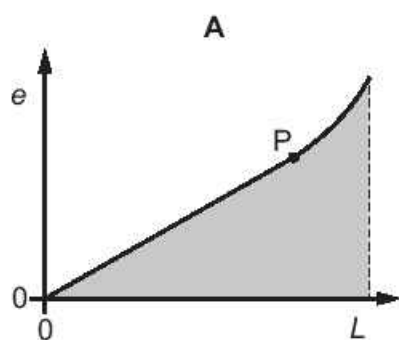
- A elastic limit
B spring constant
C stress
D Young modulus

Forces are applied to the ends of a rod so that its length increases. The variation with load L of the extension e of the rod is shown.



The point P is the elastic limit.

Which shaded area represents the work done during the plastic deformation of the rod?



139 9702/12/O/N/18/Q20

A metal cylinder is able to withstand a compressive force of 4.0 kN without deforming plastically.



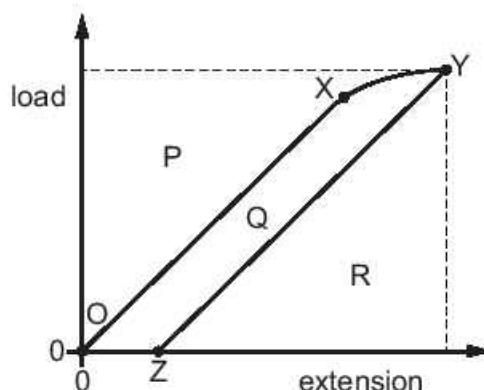
The cylinder has cross-sectional area A and would be at its elastic limit when a stress σ is applied.

What is a possible pair of values for A and σ ?

	A/m^2	σ/MPa
A	1.5×10^{-5}	50
B	1.5×10^{-5}	80
C	7.5×10^{-5}	50
D	7.5×10^{-5}	80

140 9702/12/O/N/18/Q21

A wire has both elastic and plastic properties. When it is slowly loaded, its extension varies with load as shown by line OXY. The removal of the load is represented by line YZ. This creates areas P, Q and R on the graph.

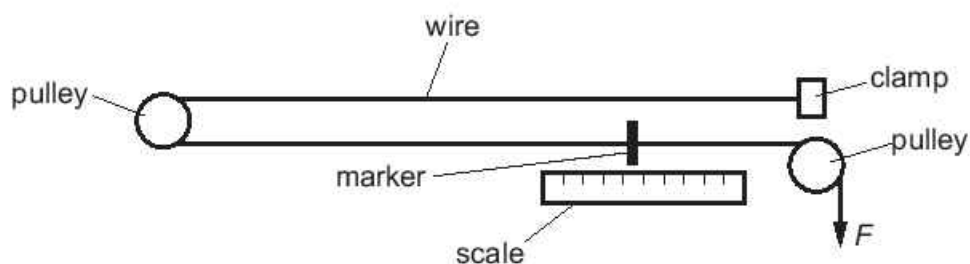


Which area represents the maximum elastic potential energy stored in the wire?

- A** P **B** Q **C** Q + R **D** R

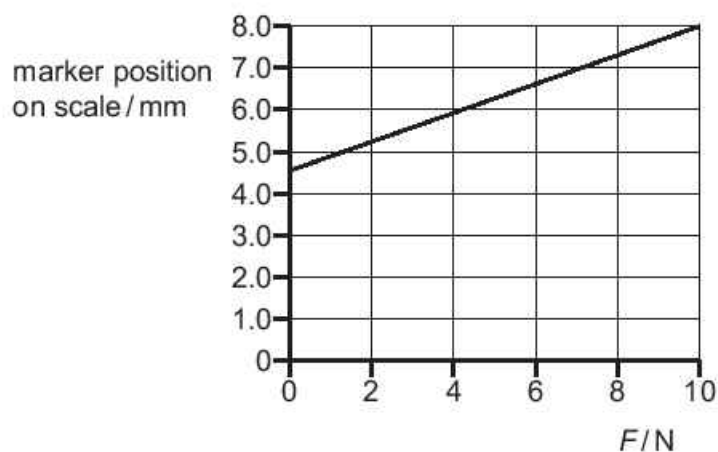
141 9702/13/O/N/18/Q19

In an experiment to measure the Young modulus of a metal, a wire of the metal of diameter 0.25 mm is clamped, as shown.



The wire passes from a clamp, around a frictionless pulley, and then to a second frictionless pulley where loads F are applied to it. A marker is attached to the wire so that the total length of wire between the clamp and the marker is initially 3.70 m. A scale is fixed near to this marker.

The graph shows how the reading on the scale varies with F .

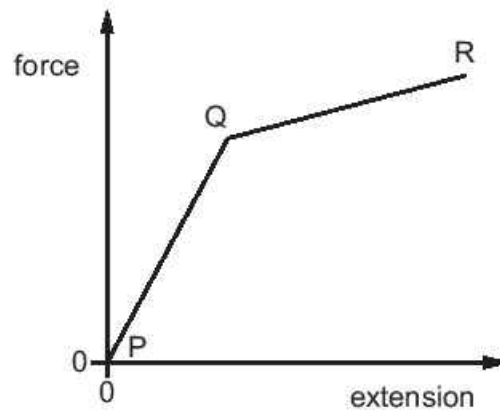


What is the Young modulus of the metal?

- A 5.5×10^{10} Pa
- B 9.4×10^{10} Pa
- C 1.6×10^{11} Pa
- D 2.2×10^{11} Pa

142 9702/13/O/N/18/Q20

A scientist is investigating the properties of a new material. She plots a force-extension graph for the material up to its breaking point.

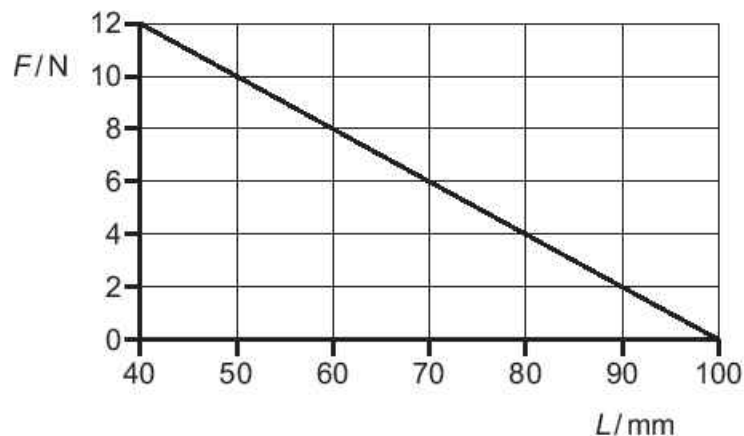


Which statement **must** be correct?

- A** The area under the graph from P to R is the strain energy stored in the material.
- B** The area under the graph from P to R is the work done in stretching the material.
- C** The material stretches elastically from Q to R.
- D** The material stretches plastically from P to Q.

143 9702/11/M/J/19/Q20

A spring of original length 100 mm is compressed by a force. The graph shows the variation of the compressing force F with the length L of the spring.

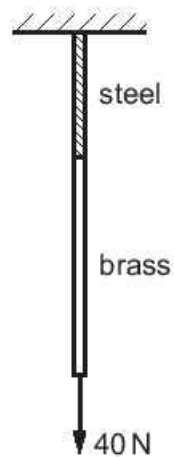


What is the energy stored in the spring when the length is 70 mm?

- A** 0.090 J **B** 0.21 J **C** 0.27 J **D** 0.63 J

144 9702/11/M/J/19/Q21

A 0.80 m length of steel wire and a 1.4 m length of brass wire are joined together. The combined wires are suspended from a fixed support and a force of 40 N is applied, as shown.



The Young modulus of steel is 2.0×10^{11} Pa.

The Young modulus of brass is 1.0×10^{11} Pa.

Each wire has a cross-sectional area of $2.4 \times 10^{-6} \text{ m}^2$.

The wires obey Hooke's law.

What is the total extension? Ignore the weights of the wires.

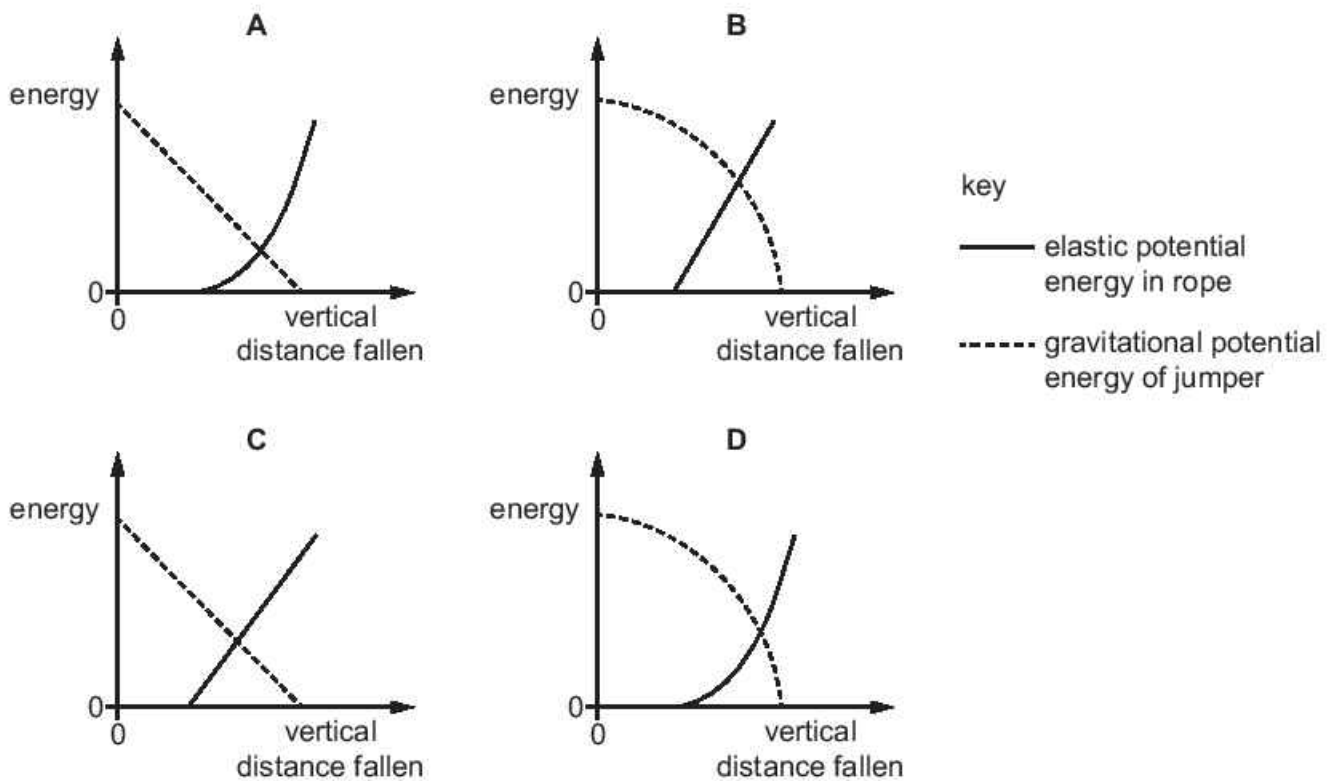
- A** $1.7 \times 10^{-4} \text{ m}$ **B** $3.0 \times 10^{-4} \text{ m}$ **C** $3.9 \times 10^{-4} \text{ m}$ **D** $9.0 \times 10^{-4} \text{ m}$

145 9702/12/M/J/19/Q18

A bungee jumper jumps off a high bridge, when attached to it by a long elastic rope which obeys Hooke's law.

The gravitational potential energy of the jumper is measured relative to the lowest point reached by the jumper.

Which graph shows the variation of the gravitational potential energy of the jumper, and the elastic potential energy in the rope, with the vertical distance fallen from the top of the bridge?



146 9702/12/M/J/19/Q20

A wire X is stretched by a force and gains elastic potential energy E .

The same force is applied to wire Y of the same material, with the same initial length but twice the diameter of wire X. Both wires obey Hooke's law.

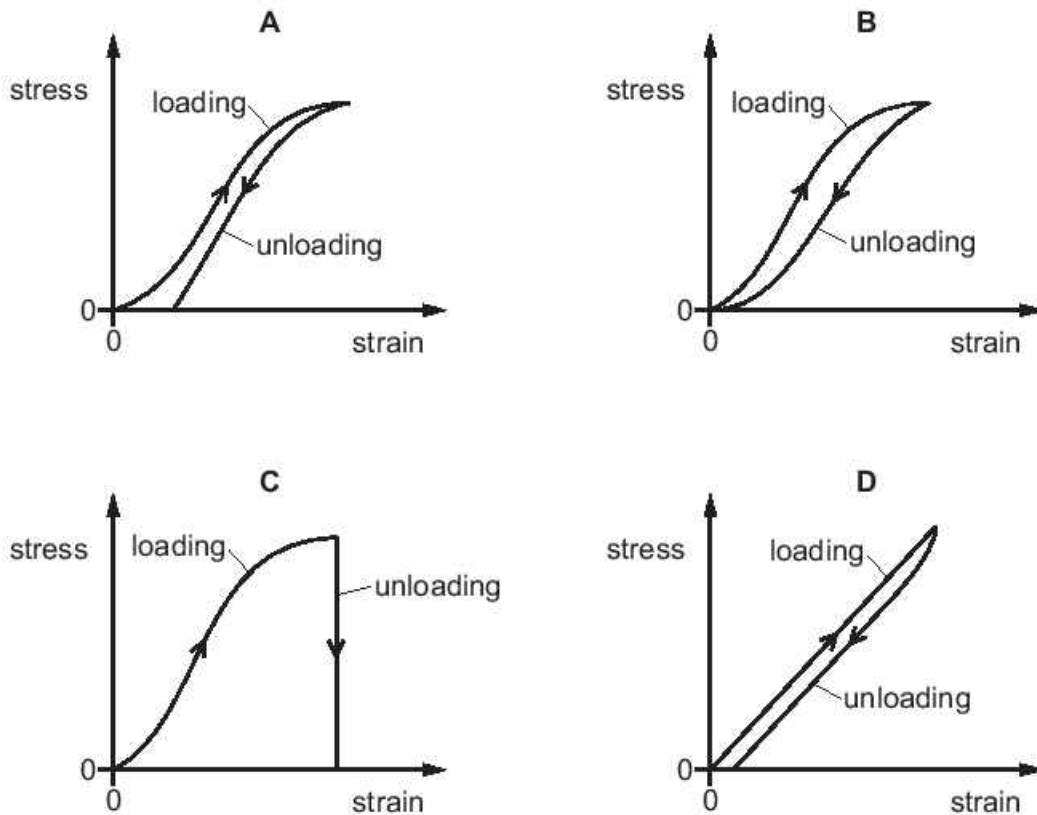
What is the gain in elastic potential energy of wire Y?

- A $0.25E$ B $0.5E$ C $2E$ D $4E$

147 9702/12/M/J/19/Q22

The stress-strain graphs for loading and unloading four different materials are shown.

Which material exhibits purely elastic behaviour?



148 9702/13/M/J/19/Q19

Four solid steel rods, each of length 2.0 m and cross-sectional area 250 mm^2 , equally support an object weighing 10 kN. The weight of the object causes the rods to contract by 0.10 mm. The rods obey Hooke's law.

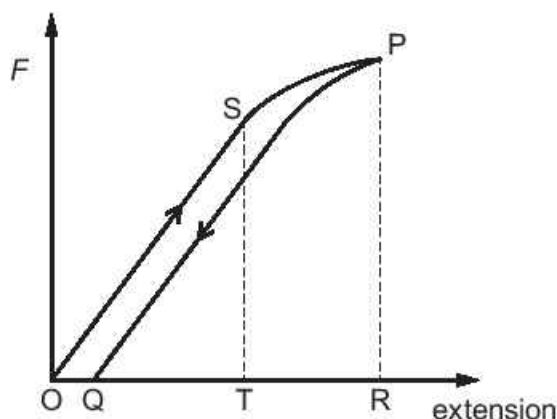
What is the Young modulus of steel?

- A $2.0 \times 10^8 \text{ Nm}^{-2}$
- B $2.0 \times 10^{11} \text{ Nm}^{-2}$
- C $8.0 \times 10^8 \text{ Nm}^{-2}$
- D $8.0 \times 10^{11} \text{ Nm}^{-2}$

149 9702/13/M/J/19/Q20

A wire is attached at one end to a fixed point. A tensile force F is applied to the other end of the wire, causing it to extend. This is shown on the graph by the line OSP.

The force F is then gradually reduced to zero and the wire contracts. This is shown on the graph by the line PQ.

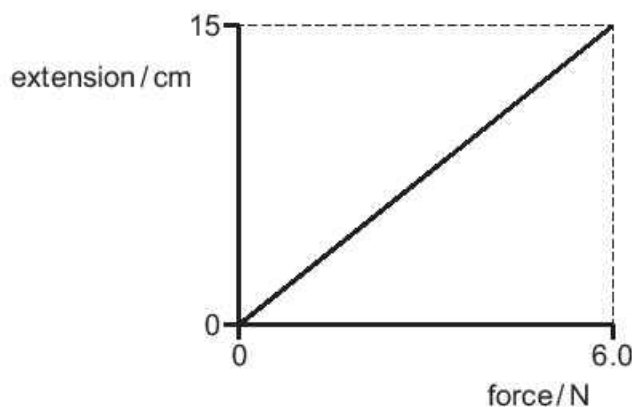


Which area on the graph represents the work done by the wire as it contracts?

- A** OSTO **B** OSPRO **C** QPRQ **D** OSPQO

150 9702/11/O/N/19/Q19

An extension–force graph for a spring is shown.

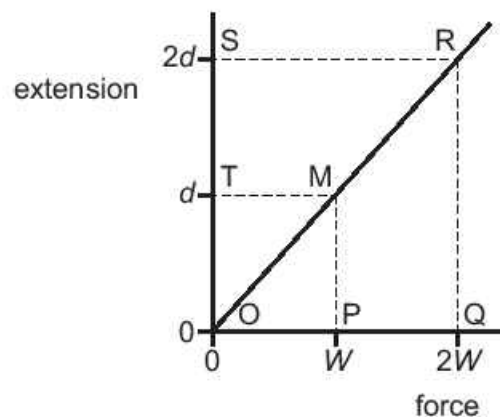


What is the spring constant of the spring?

- A** 0.025 Nm^{-1} **B** 0.40 Nm^{-1} **C** 2.5 Nm^{-1} **D** 40 Nm^{-1}

151 9702/11/O/N/19/Q20

A wire is extended by a force. The graph shows how the extension of the wire varies with the force applied.



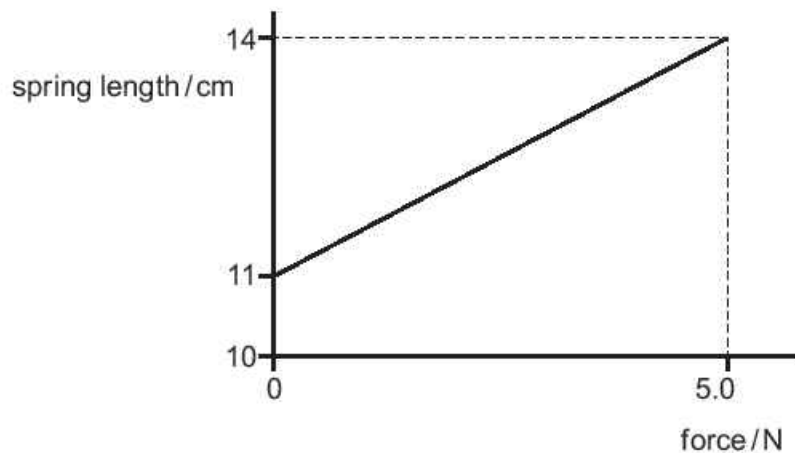
Initially a force W gives an extension d . The force is then increased to $2W$. This increases the extension to $2d$.

Which area of the graph represents the work done by the force when the force increases from W to $2W$?

- A** ORQ **B** OQRS **C** ORS **D** TMRS

152 9702/12/O/N/19/Q18

The graph shows the effect of applying a force of up to 5.0 N to a spring.



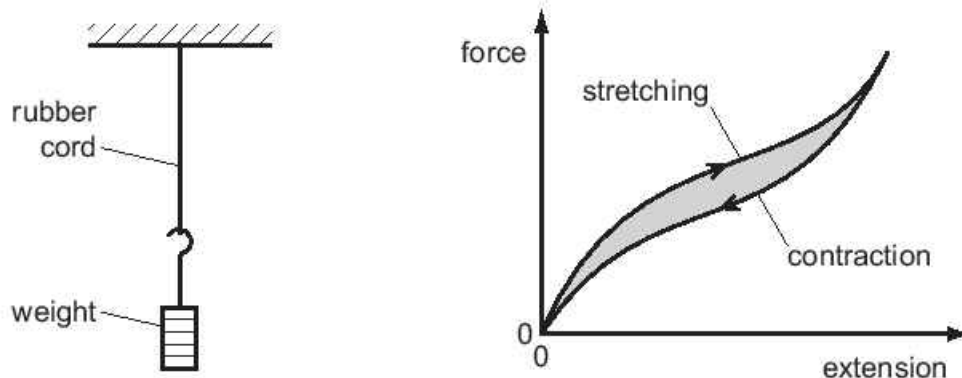
The spring obeys Hooke's law for forces up to 7.0 N.

What is the total extension of the spring produced by a 7.0 N force?

- A** 4.2 cm **B** 5.6 cm **C** 15 cm **D** 20 cm

153 9702/12/O/N/19/Q19

A rubber cord hangs from a rigid support. A weight attached to its lower end is gradually increased from zero, and then gradually reduced to zero.



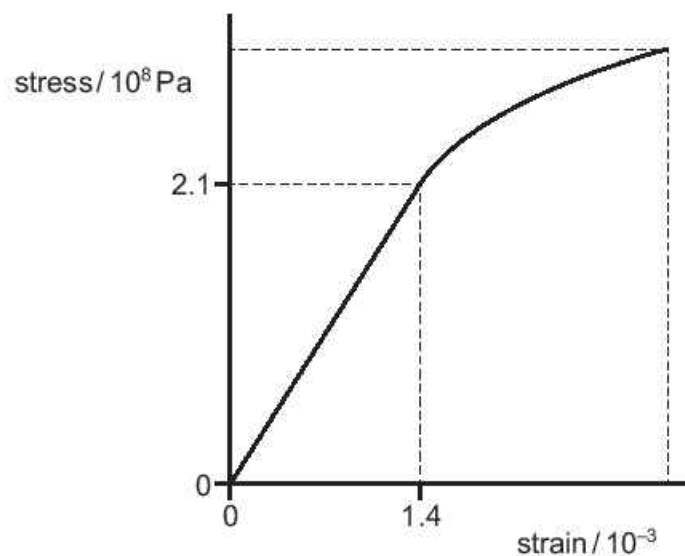
The force–extension curve for contraction is below the force–extension curve for stretching.

What does the shaded area between the curves represent?

- A the elastic potential energy stored in the rubber cord
- B the thermal energy dissipated in the rubber cord
- C the work done on the rubber cord during stretching
- D the work done by the rubber cord during contraction

154 9702/13/O/N/19/Q19

The stress–strain graph for a wire is shown.

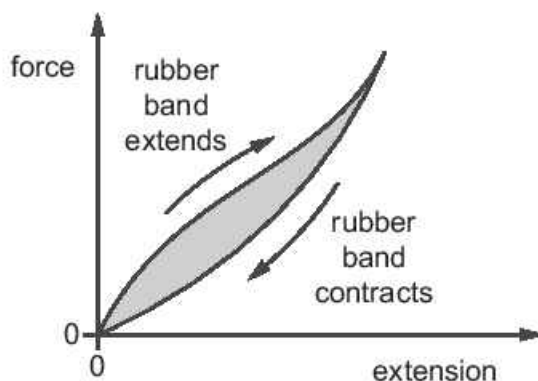


What is the Young modulus of the material of the wire?

- | | |
|------------------------------------|-----------------------------------|
| A $6.7 \times 10^{-12} \text{ Pa}$ | C $1.5 \times 10^8 \text{ Pa}$ |
| B $6.7 \times 10^{-9} \text{ Pa}$ | D $1.5 \times 10^{11} \text{ Pa}$ |

155 9702/13/O/N/19/Q20

The diagram shows a force–extension graph for a rubber band as the band is extended and then the stretching force is decreased to zero.

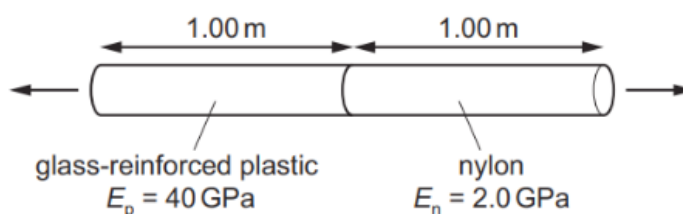


What can be deduced from the graph?

- A The rubber band does not return to its original length when the force is decreased to zero.
- B The rubber band obeys Hooke's law for the extensions shown.
- C The rubber band remains elastic for the extensions shown.
- D The shaded area represents the work done in extending the rubber band.

156 9702/12/F/M/20/Q19

A composite rod is made by attaching a glass-reinforced plastic rod and a nylon rod end to end, as shown.



The rods have the same cross-sectional area and each rod is 1.00 m in length. The Young modulus E_p of the plastic is 40 GPa and the Young modulus E_n of the nylon is 2.0 GPa.

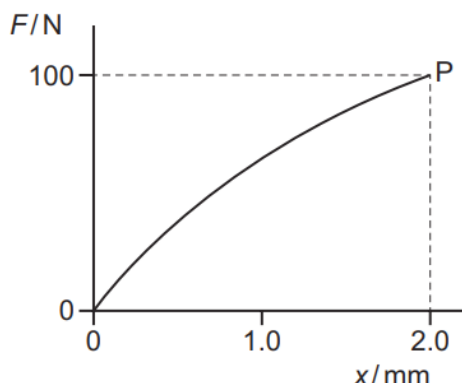
The composite rod will break when its total extension reaches 3.0 mm.

What is the greatest tensile stress that can be applied to the composite rod before it breaks?

- A $2.9 \times 10^6 \text{ Pa}$
- B $5.7 \times 10^6 \text{ Pa}$
- C $2.9 \times 10^9 \text{ Pa}$
- D $5.7 \times 10^9 \text{ Pa}$

157. 9702/12/F/M//20/Q20

The graph shows the non-linear force–extension curve for a wire made from a new composite material.



What is the best estimate of the work done in stretching the wire to point P?

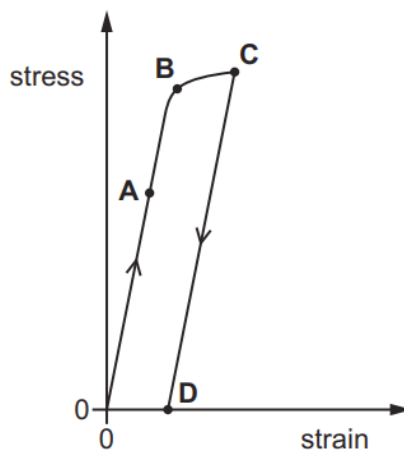
- A** 0.09 J **B** 0.10 J **C** 0.11 J **D** 0.20 J

158. 9702/11/M/J/20/Q20

A tensile force is used to extend a sample of a material. The force is then removed.

The variation with strain of the applied stress is shown on the graph.

Which point on the graph could represent the elastic limit for the material?



159. 9702/11/M/J/20/Q21

A tensile force is applied to an unstretched rubber band, causing it to stretch. The tensile force is then removed.

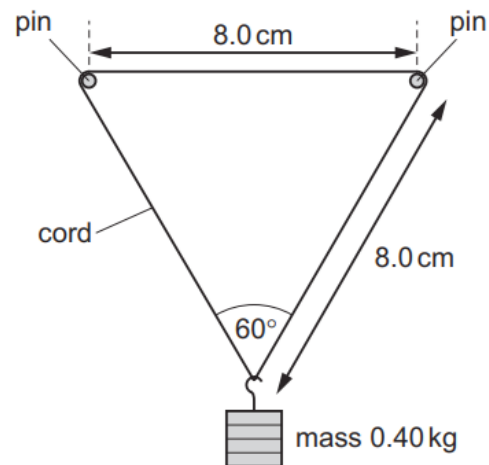
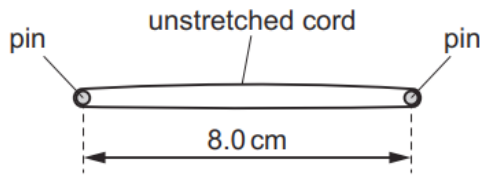
Which statement about the rubber band **must** be correct?

- A** If the rubber band stretches elastically and plastically, all the work done by the force is converted to thermal energy in the rubber.
- B** If the rubber band stretches elastically, it obeys Hooke's law.
- C** If the rubber band stretches elastically, the gradient of the force–extension graph represents the work done by the force.
- D** If the rubber band stretches plastically, the rubber band will be longer after the force is removed than it was before the force is applied.

160. 9702/12/M/J/20/Q18

An elastic cord of unstretched total length 16.0 cm and cross-sectional area $2.0 \times 10^{-6}\text{ m}^2$ is held horizontally by two smooth pins a distance 8.0 cm apart.

The cord obeys Hooke's law. A load of mass 0.40 kg is suspended centrally on the cord. The angle between the two sides of the cord supporting the load is 60° .



What is the Young modulus of the cord material?

- A** $5.7 \times 10^5\text{ Pa}$ **B** $1.1 \times 10^6\text{ Pa}$ **C** $2.3 \times 10^6\text{ Pa}$ **D** $3.9 \times 10^6\text{ Pa}$

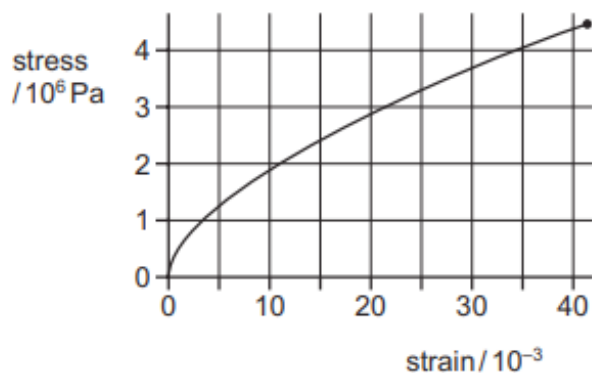
161. 9702/12/M/J/20/Q20

Which statement describes what is meant by the plastic deformation of a material?

- A** It always obeys Hooke's law.
- B** It does not return to its original length when the extending force is removed.
- C** It never obeys Hooke's law.
- D** It returns to its original length when the extending force is removed.

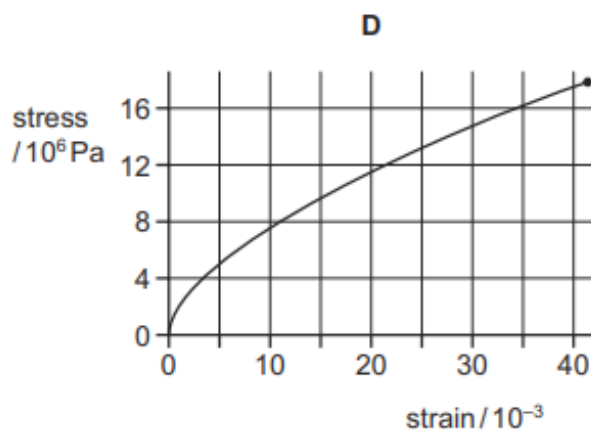
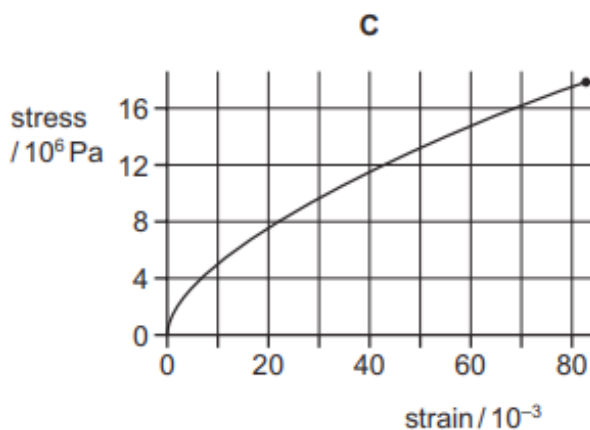
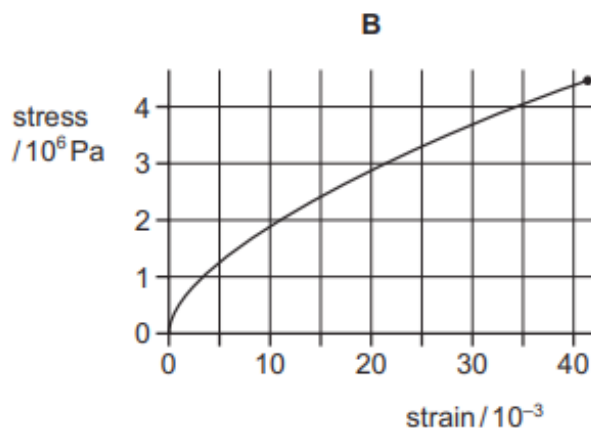
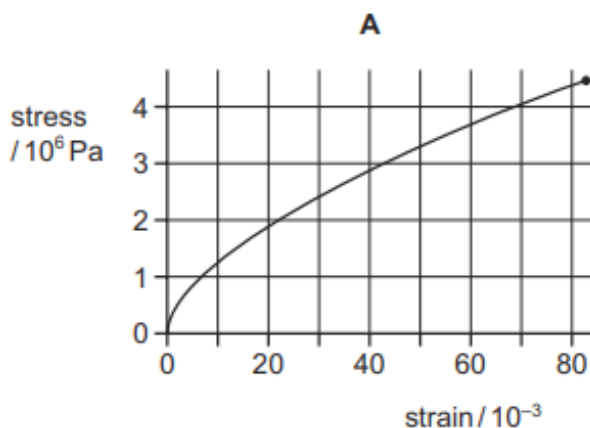
162. 9702/12/M/J/20/Q19

A student is investigating the mechanical properties of a metal. He applies different loads to a long thin wire up to its breaking point, and measures the extension of the wire for each load. He then plots a graph of stress against strain.



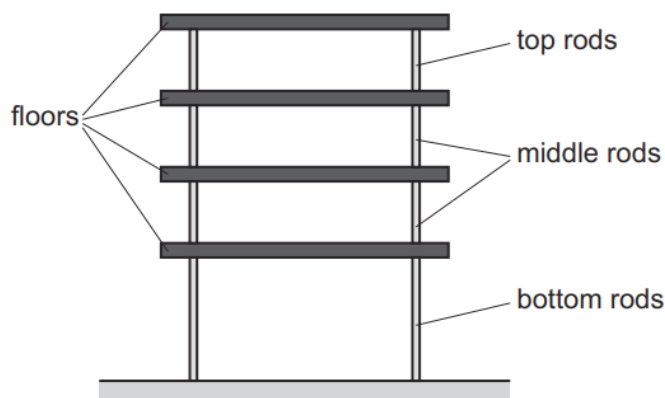
The student repeats the experiment with a wire made from the same metal, with twice the original length and half the diameter.

Which graph is obtained?



163. 9702/13/M/J/20/Q20

The diagram shows a simplified model of a building with four identical heavy floors.



The spacing of the bottom floor from the ground is twice that of the spacing between the floors. Between each floor are equal numbers of vertical steel supporting rods of negligible mass compared with the floors. The rods are of different diameters so that the stress in each rod is the same.

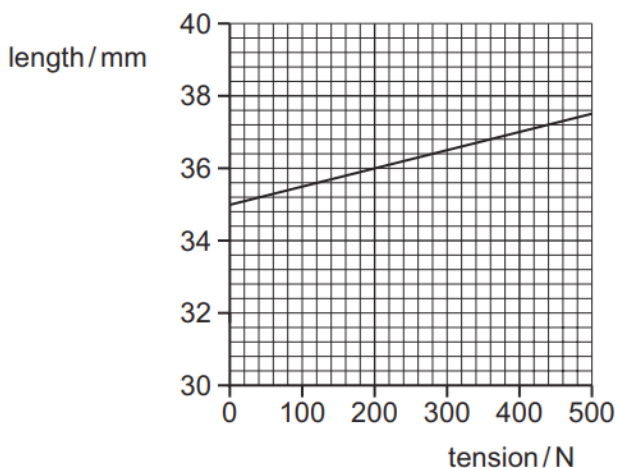
What is the ratio $\frac{\text{diameter of bottom rods}}{\text{diameter of top rods}}$?

- A** 2 **B** 4 **C** 8 **D** 16

164. 9702/13/M/J/20/Q21

The Achilles tendon in a rabbit's leg is stretched when the rabbit jumps.

The graph shows the variation with tension of the length of the tendon.

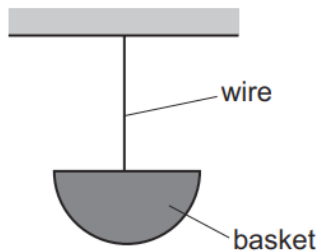


What is the strain energy in the tendon when the tension is 400 N?

- A** 0.40 J **B** 0.80 J **C** 2.4 J **D** 7.4 J

165. 9702/11/O/N/20/Q19

A wire of circular cross-section, which obeys Hooke's law, is used to suspend a basket as shown.



The Young modulus for the material of the wire is $2.5 \times 10^{11} \text{ Pa}$.

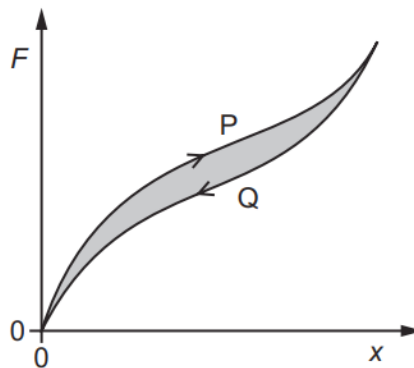
When a weight of 34 N is added to the basket, the strain in the wire increases by 6.0×10^{-5} .

What is the radius of the wire?

- A** $7.2 \times 10^{-7} \text{ m}$ **B** $2.3 \times 10^{-6} \text{ m}$ **C** $8.5 \times 10^{-4} \text{ m}$ **D** $1.7 \times 10^{-3} \text{ m}$

166. 9702/11/O/N/20/Q20

An unstretched rubber cord is stretched by a force. The force F is plotted against the extension x . F is slowly increased from zero, causing the cord to extend along path P. F is then reduced back to zero along path Q.



What is represented by the shaded area?

- A** the elastic energy stored in the rubber cord
B the energy that causes plastic deformation
C the energy dissipated as heat
D the work done to extend the rubber cord

167. 9702/12/O/N/20/Q20

A mass of 60.0 g is suspended from a spring and the distance from the bottom of the spring to the floor is measured to be 16.4 cm.

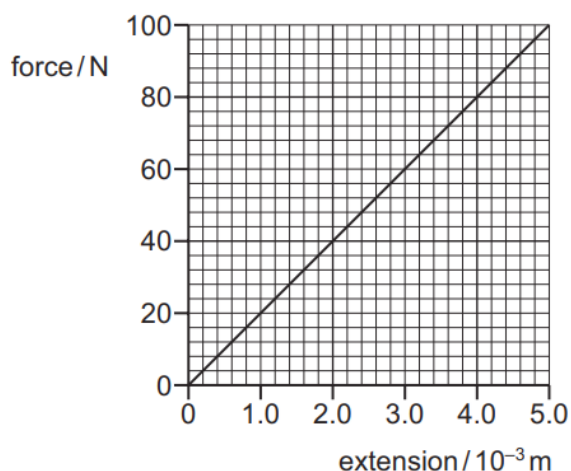
The mass is replaced with a 100.0 g mass and the distance from the bottom of the spring to the floor is now measured to be 12.6 cm. The spring obeys Hooke's law.

What is the spring constant of the spring?

- A** 1.05 N m^{-1} **B** 1.35 N m^{-1} **C** 10.3 N m^{-1} **D** 103 N m^{-1}

168. 9702/12/O/N/20/Q21

The graph shows the force–extension graph for a wire.



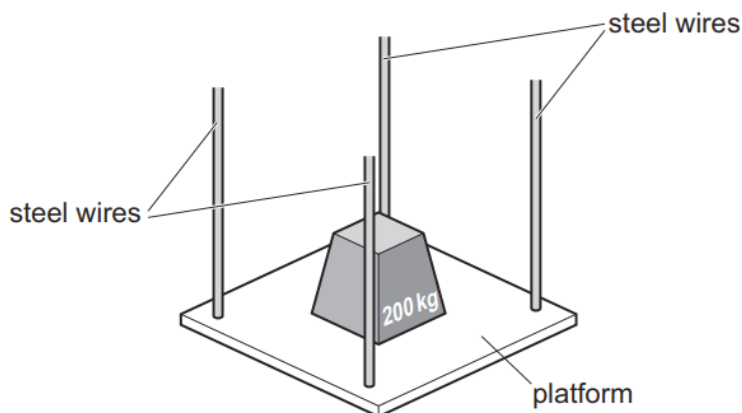
The wire is already extended by a force of 60 N.

How much work is done to increase the extension of the wire by 2.0 mm?

- A** 0.040 J **B** 0.090 J **C** 0.16 J **D** 0.25 J

169. 9702/13/O/N/20/Q20

A platform is suspended by four steel wires. Each wire is 5.0 m long and has a diameter of 3.0 mm. The Young modulus of steel is $2.1 \times 10^{11} \text{ Pa}$.



The wires obey Hooke's law when a load of mass 200 kg is placed on the platform.

How far will the platform descend because of the extension of the wires?

- A** $1.7 \times 10^{-4} \text{ m}$ **B** $4.1 \times 10^{-4} \text{ m}$ **C** $1.7 \times 10^{-3} \text{ m}$ **D** $6.6 \times 10^{-3} \text{ m}$

170. 9702/13/O/N/20/Q21

A tensile force of 7.00 MN is applied to a sample of steel. This causes the sample to extend by 5.00 mm in the direction of the force. The sample obeys Hooke's law.

What is the work done to extend the sample?

- A** 17.5 J **B** 35.0 J **C** 17.5 kJ **D** 35.0 kJ

171. 9702/12/F/M/21/Q19

Which expression is equal to the stress on a wire?

- A** $\frac{\text{extension}}{\text{original length}}$
- B** $\frac{\text{force}}{\text{cross-sectional area}}$
- C** $\frac{\text{force}}{\text{extension}}$
- D** $\frac{\text{Young modulus}}{\text{original length}}$

172. 9702/12/F/M/21/Q20

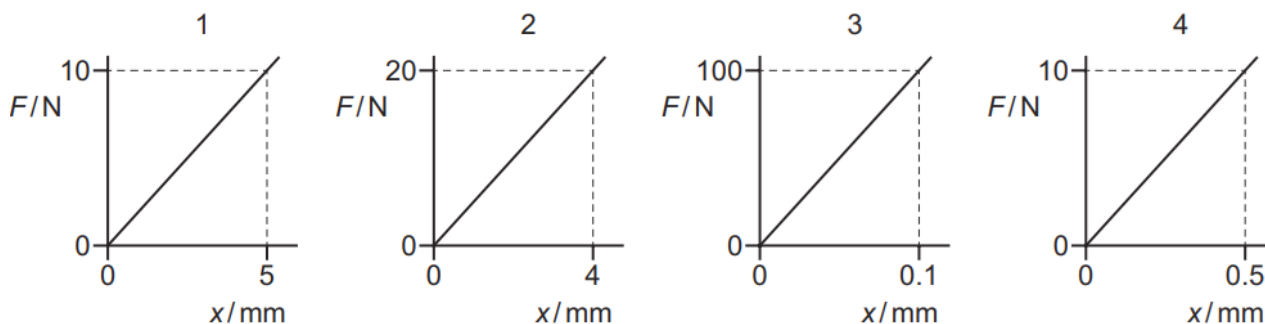
A wire is stretched by applying increasing values of force F . For each value of force applied, the extension x is recorded. A force–extension graph is plotted from the data obtained.

Which statement about the area under the graph **must** be correct?

- A** It can be calculated as $\frac{1}{2}Fx$.
- B** It is the elastic potential energy stored in the stretched sample.
- C** It is the work done in stretching the sample.
- D** It would be the same for any wire of the same material.

173. 9702/11/M/J/21/O19

The spring constants of four springs are determined by plotting the following graphs of force F against extension x .

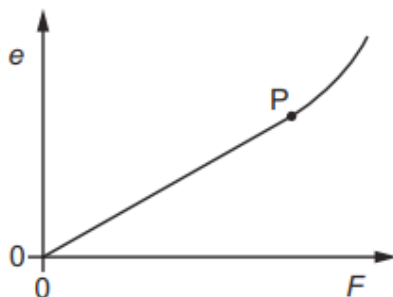


Which order of the graphs shows **decreasing** spring constants?

- A** $2 \rightarrow 1 \rightarrow 3 \rightarrow 4$
- B** $3 \rightarrow 4 \rightarrow 2 \rightarrow 1$
- C** $4 \rightarrow 2 \rightarrow 1 \rightarrow 3$
- D** $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

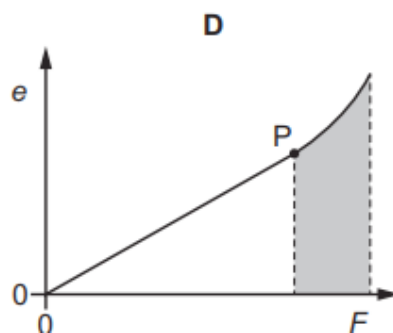
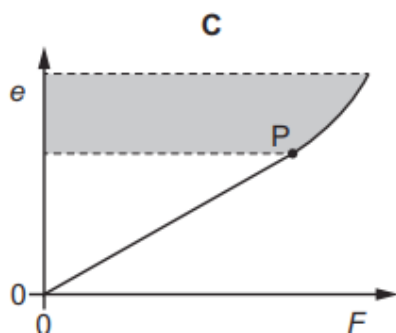
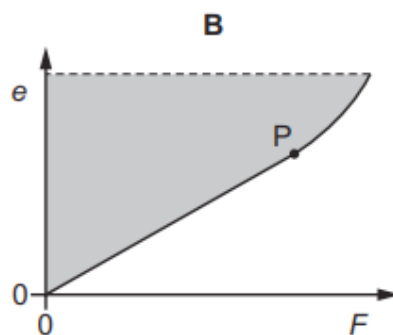
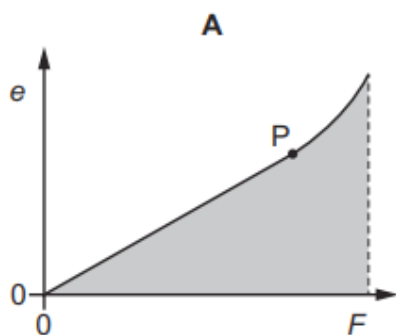
174. 9702/11/M/J/21/Q20

Forces are applied to the ends of a rod so that its length increases. The variation with force F of the extension e of the rod is shown.



The point P is the elastic limit.

Which shaded area represents the work done during the plastic deformation of the rod?



175. 9702/12/M/J/21/Q19

A steel bar of circular cross-section is under tension T , as shown.

The diameter of the wide portion is double the diameter of the narrow portion.



What is the value of $\frac{\text{stress in the wide portion}}{\text{stress in the narrow portion}}$?

A 0.25

B 0.50

C 2.0

D 4.0

176. 9702/12/M/J/21/Q20

Two guitar strings are stretched by tensile forces.

String X is stretched by a tensile force F that causes an extension x .

String Y is stretched by a tensile force $2F$ that causes an extension $2x$.

The strings obey Hooke's law.

What is the ratio $\frac{\text{strain energy in stretched string X}}{\text{strain energy in stretched string Y}}$?

A 4

B 2

C $\frac{1}{2}$

D $\frac{1}{4}$

177. 9702/13/M/J/21/Q16

A spring is initially neither compressed nor extended.

A force can be applied to this spring so that it is either compressed to a shorter length or extended to a longer length.

What is the change in the elastic potential energy in the spring when it is extended and when it is compressed?

	change in the elastic potential energy	
	spring is extended	spring is compressed
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

178. 9702/13/M/J/21/Q20

The stress σ in a material is given by the equation shown.

$$\sigma = \frac{F}{A}$$

The strain ε in the same material is given by the equation shown.

$$\varepsilon = \frac{x}{L}$$

Which expression gives the Young modulus of the material?

A $\frac{\varepsilon}{\sigma}$

B $\frac{Fx}{AL}$

C $\frac{\sigma x}{L}$

D $\frac{F}{A\varepsilon}$

179. 9702/13/M/J/21/Q21

What is an example of plastic deformation?

- A** A rubber ball is momentarily compressed every time it hits the ground.
- B** A spoon stirring some coffee in a ceramic mug hits its surface and makes a clinking sound.
- C** A toolbox is left on a horizontal plank. When the toolbox is removed, the plank is no longer straight.
- D** The spring in some bathroom weighing scales is compressed by a person standing on the scales.

180. 9702/11/O/N/21/Q19

A metal wire, of cross-sectional area A and unstretched length l , is subjected to stress σ . As a result it has strain ε .

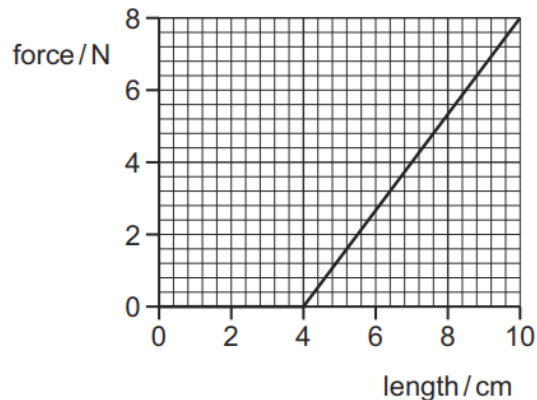
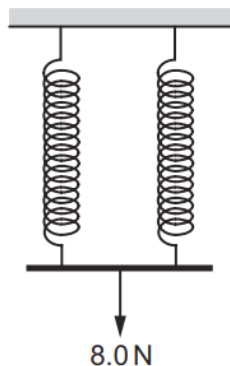
Which expression gives the Young modulus of the metal?

- A** $\frac{\varepsilon}{\sigma}$ **B** $\frac{\varepsilon A}{\sigma l}$ **C** $\frac{\sigma}{\varepsilon}$ **D** $\frac{\sigma l}{\varepsilon A}$

181. 9702/11/O/N/21/Q20

Two identical springs are connected in parallel.

A weight of 8.0 N is hung from the combination, as shown.



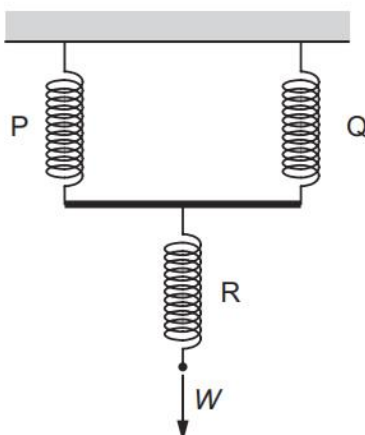
The graph shows the variation with length of the force applied to **one** of the springs.

What is the strain energy in **one** of the springs?

- A** 0.060 J **B** 0.12 J **C** 0.14 J **D** 0.24 J

182. 9702/12/O/N/21/Q21

Three springs are arranged vertically as shown.



Springs P and Q are identical and each has spring constant k . Spring R has spring constant $3k$.

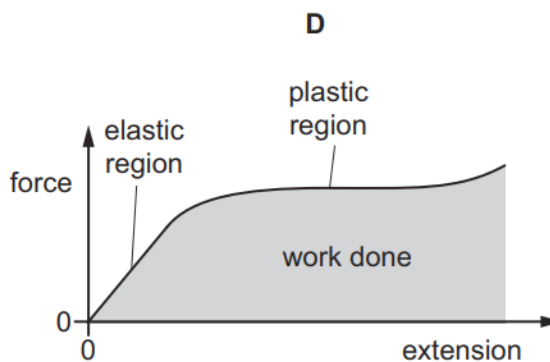
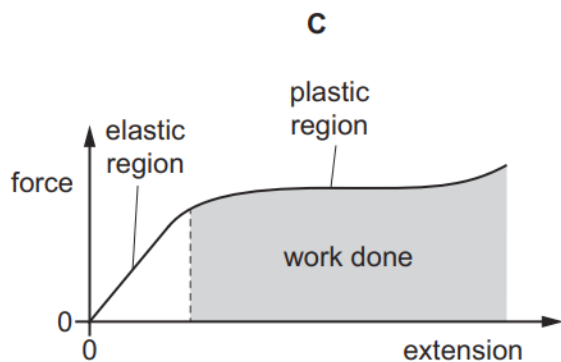
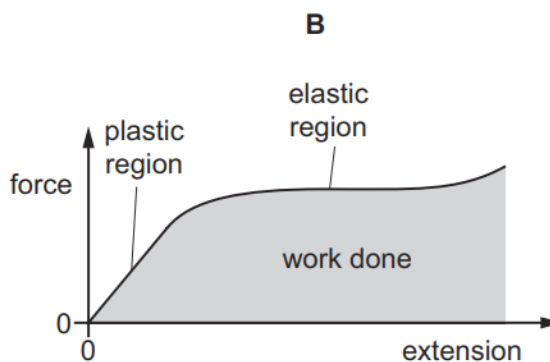
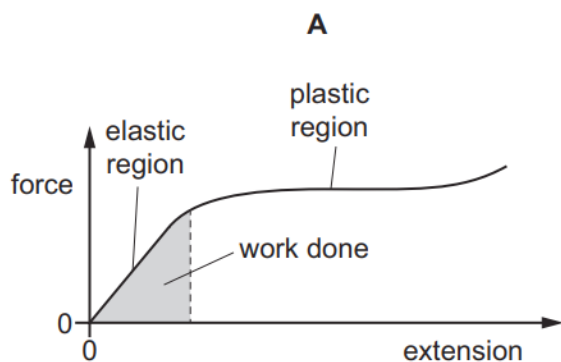
What is the increase in the overall length of the arrangement when a force W is applied as shown?

- A** $\frac{5W}{6k}$ **B** $\frac{4W}{3k}$ **C** $\frac{7}{2}kW$ **D** $4kW$

183. 9702/13/O/N/21/Q21

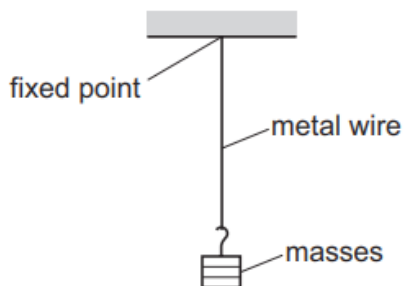
A metal wire is stretched to breaking point and the force—extension graph is plotted.

Which graph is correctly labelled with the elastic region, the plastic region and the area representing the work done to stretch the wire until it breaks?

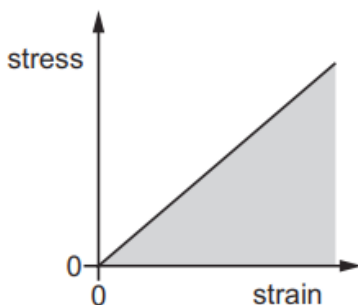


184. 9702/12/O/N/21/Q22

A length of metal wire is attached to a fixed point and hangs vertically. Masses are then suspended from the wire. Assume that the cross-sectional area of the wire remains constant.



A stress–strain graph for the wire is plotted, as shown.

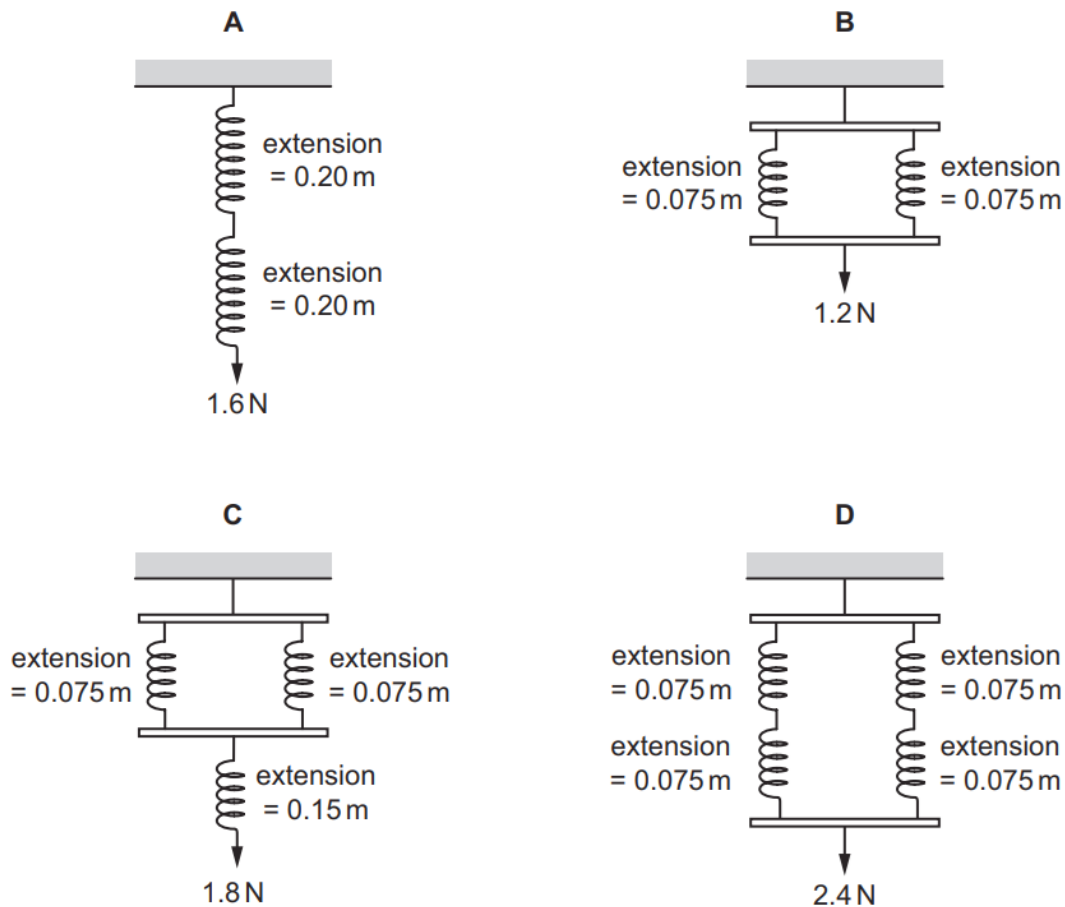


What is represented by the shaded area under the graph?

- A strain energy in the wire
- B $\frac{\text{strain energy in the wire}}{\text{cross-sectional area of the wire}}$
- C $\frac{\text{strain energy in the wire}}{\text{original length of the wire}}$
- D $\frac{\text{strain energy in the wire}}{\text{original volume of the wire}}$

185. 9702/13/O/N/21/Q20

Which spring combination has an overall spring constant of 8.0 N m^{-1} ?



186. 9702/12/F/M/22/Q18

A metal wire is stretched. The wire obeys Hooke's law.

Which quantity has a value that does **not** change?

- A** extension
- B** strain
- C** stress
- D** Young modulus

187. 9702/12/F/M/22/Q19

An object is stretched until it reaches the elastic limit.

Which statement must describe the stress on the object when it is at the elastic limit?

- A** It is the maximum stress for which the object obeys Hooke's law.
- B** It is the maximum stress that can be applied to the object before it has elastic deformation.
- C** It is the maximum stress that can be applied to the object before it has plastic deformation.
- D** It is the maximum stress the object can withstand before it breaks.

188. 9702/11/M/J/22/Q19

Two wires, P and Q, are made from the same metal and hang vertically from a steel girder.

Wire Q has half the length and twice the diameter of wire P.

Identical masses are attached to the bottom of each wire. Both wires obey Hooke's law as they are stretched by the weight of the masses.

What is the ratio $\frac{\text{extension of wire P}}{\text{extension of wire Q}}$?

A $\frac{8}{1}$

B $\frac{4}{1}$

C $\frac{1}{1}$

D $\frac{1}{2}$

189. 9702/11/M/J/22/Q20

Which statement about elastic and plastic deformation **must** be correct?

- A** Elastic deformation and plastic deformation are proportional to the applied force.
- B** Elastic deformation and plastic deformation cause no change in volume.
- C** Elastic deformation causes heating of the material but plastic deformation does not.
- D** Elastic deformation is reversible but plastic deformation is not.

190. 9702/12/M/J/22/Q19

A metal wire obeys Hooke's law and has a Young modulus of 2.0×10^{11} Pa. The wire has an original length of 1.6 m and a diameter of 0.48×10^{-3} m.

What is the spring constant of the wire?

A $7.2 \times 10^3 \text{ N m}^{-1}$

B $2.3 \times 10^4 \text{ N m}^{-1}$

C $2.9 \times 10^4 \text{ N m}^{-1}$

D $9.0 \times 10^4 \text{ N m}^{-1}$

191. 9702/12/M/J/22/Q20

A wire is being stretched by a tensile force.

Which statement about the elastic limit **must** be correct?

- A** The deformation is plastic after the elastic limit has been reached.
- B** The deformation is plastic until the elastic limit is reached.
- C** The extension is proportional to the tensile force after the elastic limit has been reached.
- D** The extension is proportional to the tensile force until the elastic limit is reached.

192. 9702/13/M/J/22/Q20

A steel wire has a length of 300 cm and a cross-sectional area of 0.50 mm^2 . The Young modulus of steel is $2.0 \times 10^{11} \text{ Pa}$.

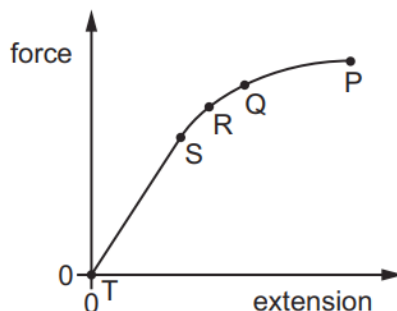
One end of the wire is attached to a fixed point. A load of 10 N is hung from the other end. The wire obeys Hooke's law.

What is the extension of the wire?

- A** $3.0 \times 10^{-7} \text{ m}$
- B** $3.0 \times 10^{-5} \text{ m}$
- C** $3.0 \times 10^{-4} \text{ m}$
- D** $3.0 \times 10^{-2} \text{ m}$

193. 9702/13/M/J/22/Q21

The extension of a copper wire is measured for different forces applied to the wire. A graph is plotted to show the variation of the force on the wire against extension. The maximum force is applied at point P.



Which statement **must** be correct?

- A** Point R is the limit of proportionality.
- B** The elastic potential energy of the wire at point S is given by the area under the graph between points T and S.
- C** There is no plastic deformation between points Q and P.
- D** The wire obeys Hooke's law up to a point between R and Q.

194. 9702/11/O/N/22/O20

An unstretched spring has a length of 2.0 cm.

The spring is then stretched within its limit of proportionality by a tensile force of 1.5 N so that the elastic potential energy stored in the spring is 0.045 J.

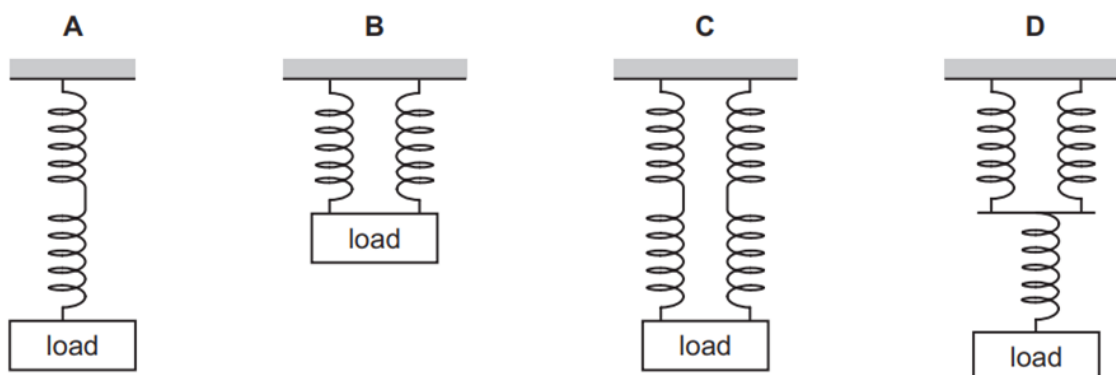
What is the stretched length of the spring?

- A** 3.0 cm
- B** 5.0 cm
- C** 6.0 cm
- D** 8.0 cm

195. 9702/11/O/N/22/Q21

Identical springs are joined in four arrangements.

Which arrangement has the same spring constant as a single spring?



196. 9702/12/O/N/22/Q20

A known tensile force acts on a metal wire. The wire does not exceed its limit of proportionality.

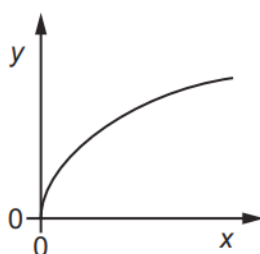
Which two measurements enable the strain of the wire to be calculated?

- A** the unstretched length of the wire and the cross-sectional area of the wire
- B** the unstretched length of the wire and the extension of the wire
- C** the Young modulus of the metal and the extension of the wire
- D** the Young modulus of the metal and the unstretched length of the wire

197. 9702/12/O/N/22/Q21

A wire is extended by different forces. The wire obeys Hooke's law.

A graph is plotted to show the variation of a quantity y with a quantity x .



What could x and y represent?

	x	y
A	elastic potential energy	extension
B	extension	force
C	force	extension
D	extension	elastic potential energy

198. 9702/13/O/N/22/Q19

A copper wire of length 3.6 m and diameter 1.22 mm is stretched by a force of 37 N. The wire obeys Hooke's law. The Young modulus of copper is 1.17×10^{11} Pa.

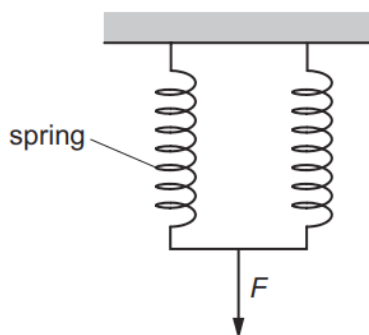
Which extension is caused by this force?

- A** 0.24 mm **B** 0.76 mm **C** 0.97 mm **D** 3.1 mm

199. 9702/13/O/N/22/Q20

A spring has spring constant k . The spring obeys Hooke's law and experiences extension x when a force F is applied to it. The resulting elastic potential energy of the spring is E_P .

The diagram shows two of these springs joined together in parallel and hanging from a fixed beam.



What is the extension and total elastic potential energy of this arrangement when the same force F is applied?

	extension	total elastic potential energy
A	$\frac{1}{2}x$	$\frac{1}{2}E_P$
B	$\frac{1}{2}x$	$\frac{1}{4}E_P$
C	x	$\frac{1}{2}E_P$
D	x	$\frac{1}{4}E_P$

200. 9702/12/F/M/23/Q18

A spring has an unstretched length of 4.50 cm. The spring is fixed at one end and a force of 35.0 N is applied to the other end so that the spring extends.

The spring obeys Hooke's law and has a spring constant of 420 N m^{-1} .

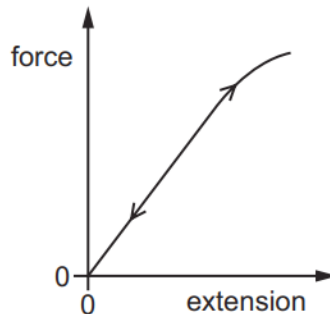
What is the strain of the extended spring?

- A** 0.019 **B** 0.083 **C** 1.85 **D** 2.67

201. 9702/12/F/M/23/Q19

A wire is fixed at one end and extended by a force that is applied to the other end. The force is slowly increased from zero and then slowly decreased back to zero.

The force–extension graph for the wire is shown.



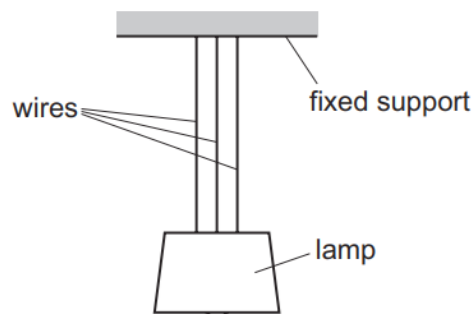
The graph line for the wire being loaded is the same as the graph line for the wire being unloaded.

Which statement describes the deformation of the wire?

- A** It is both elastic and plastic.
- B** It is elastic only.
- C** It is neither elastic nor plastic.
- D** It is plastic only.

202. 9702/11/M/J/23/Q18

A lamp is suspended in equilibrium from a fixed support by three long identical wires.



The weight of the lamp causes each wire to have an extension of 0.40 cm. The height h of the lamp above the floor is measured.

The middle wire suddenly breaks and the lamp falls a small distance as the extensions of the remaining two wires increase. The wires obey Hooke's law.

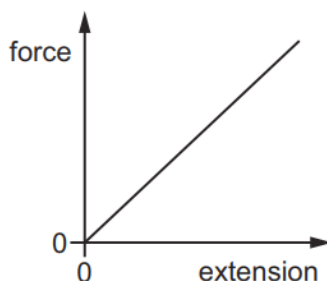
When the lamp is in equilibrium, the height h of the lamp above the floor is measured again.

What is the difference between the two values of h ?

- A** 0.20 cm
- B** 0.27 cm
- C** 0.40 cm
- D** 0.60 cm

203. 9702/11/M/J/23/Q19

The force–extension graph for a spring is shown.



What represents the work done to extend the spring?

- A** the area under the graph
- B** the gradient of the graph
- C** the reciprocal of the gradient of the graph
- D** twice the area under the graph

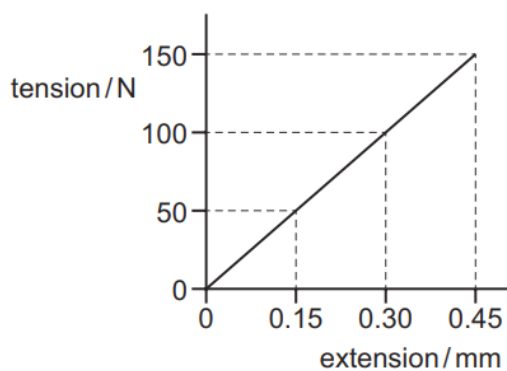
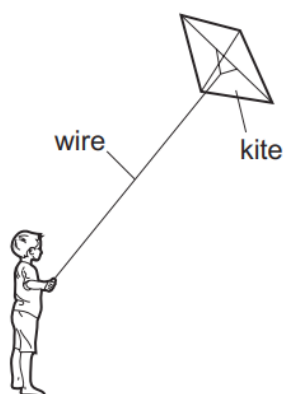
204. 9702/12/M/J/23/Q18

What is meant by the spring constant of a spring?

- A** extension per unit force
- B** $\frac{1}{2} \times \text{force} \times \text{extension}$
- C** force $\times$ extension
- D** force per unit extension

205. 9702/12/M/J/23/Q19

A child holds a thin metal wire that is attached to a kite. The graph shows how the extension of the wire varies with its tension.



A gust of wind increases the tension from 100 N to 150 N.

What is the change in the elastic potential energy of the wire caused by the gust of wind?

- A** 3.8 mJ
- B** 19 mJ
- C** 34 mJ
- D** 38 mJ

206. 9702/13/M/J/23/Q18

A metal wire has length 5.2 m and diameter 1.0 mm. The metal has Young modulus 360 GPa.

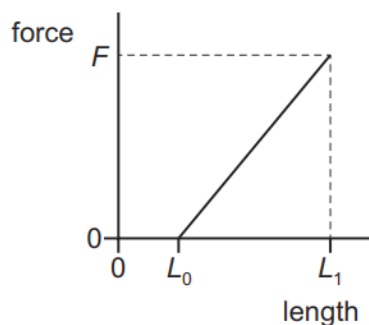
The wire is fixed at one end and a force is applied to the other end. The force extends the wire by 7.2 mm. The wire obeys Hooke's law.

What is the force applied to the wire?

- A** $1.2 \times 10^2 \text{ N}$
- B** $3.9 \times 10^2 \text{ N}$
- C** $5.0 \times 10^2 \text{ N}$
- D** $1.6 \times 10^3 \text{ N}$

207. 9702/13/M/J/23/Q19

The graph shows how the length of a spring varies with the force applied to it.



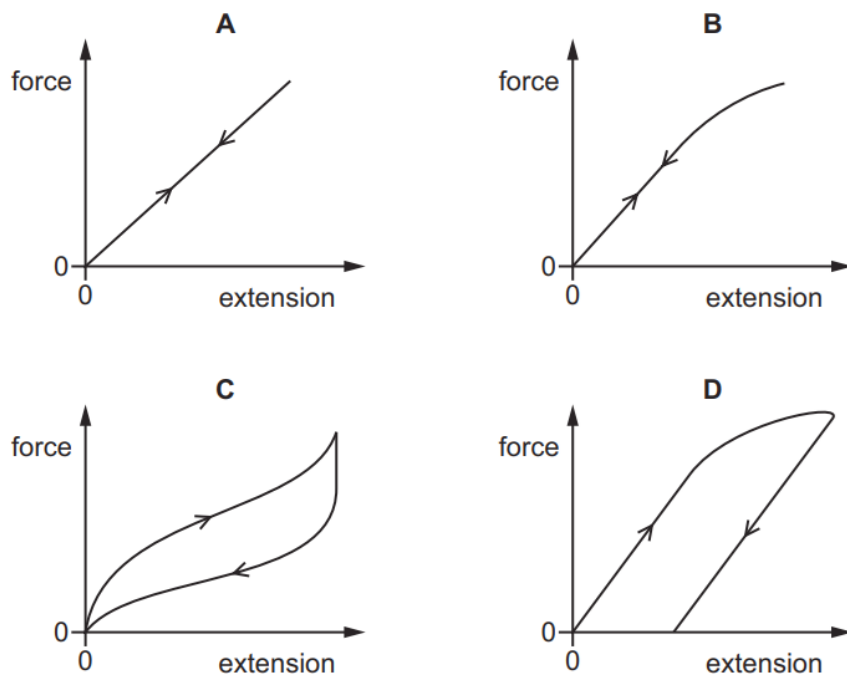
The spring has unstretched length L_0 . When a force F is applied, the spring has length L_1 .

What is the work done in stretching the spring to length L_1 ?

- A** $\frac{1}{2} FL_1$
- B** $\frac{1}{2} F(L_1 - L_0)$
- C** FL_1
- D** $F(L_1 - L_0)$

208. 9702/11/O/N/23/Q19

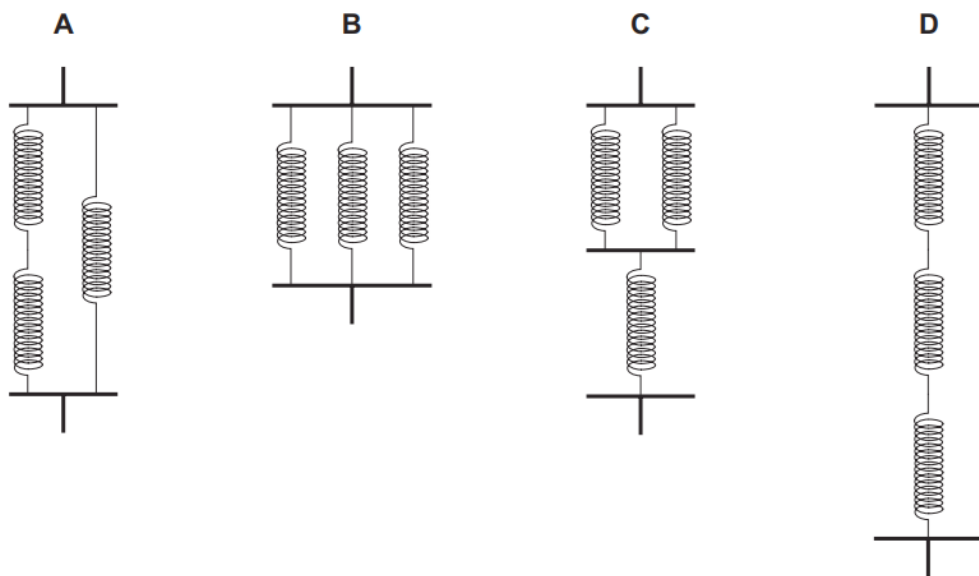
Which force–extension graph shows plastic deformation of a sample of material?



209. 9702/12/O/N/23/Q18

Three identical springs, each with the same spring constant, are connected together in four different arrangements, as shown.

Which arrangement has the largest combined spring constant?



210. 9702/13/O/N/23/Q18

A copper wire of diameter 1.6 mm is stretched within its limit of proportionality by a tensile force of 430 N.

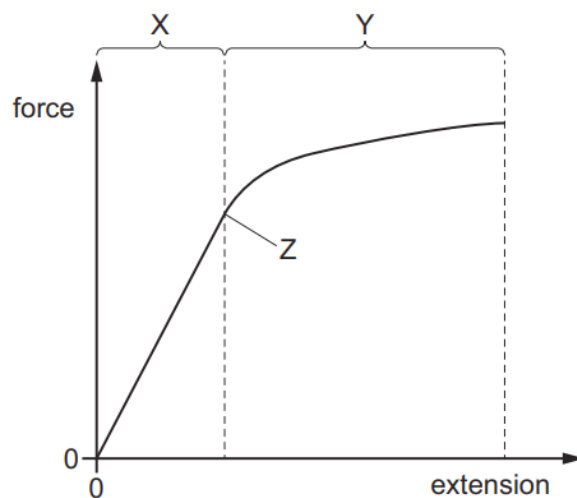
The Young modulus of copper is 130 GPa.

What is the strain in the wire?

- A** 4.1×10^{-4} **B** 1.3×10^{-3} **C** 1.6×10^{-3} **D** 5.2×10^{-3}

211. 9702/12/O/N/23/Q19

The force–extension graph for a wire is shown.

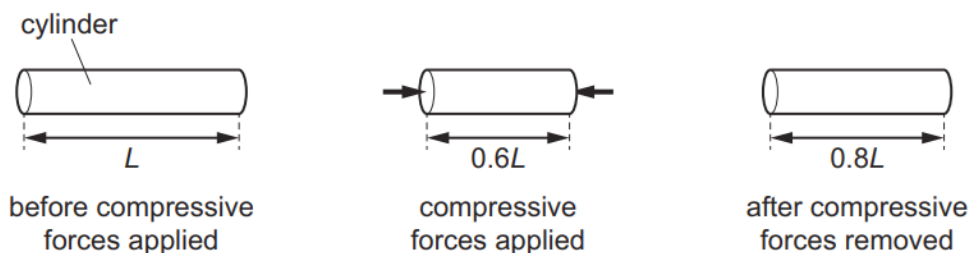


Which row could identify the labels X, Y and Z?

	limit of proportionality	region of elastic deformation	region of plastic deformation
A	X	Y	Z
B	Z	Y	X
C	Y	Z	X
D	Z	X	Y

212. 9702/13/O/N/23/Q19

Compressive forces are applied normally to the end faces of a cylinder of initial length L . The cylinder is compressed by the forces so that its length decreases to $0.6L$. After the compressive forces are removed, the cylinder's length increases to $0.8L$.



What describes the deformation of the cylinder when its length was $0.6L$?

- A** both elastic and plastic
- B** elastic only
- C** plastic only
- D** neither elastic nor plastic